

4

Differential Equations

4.1. Differential Equations

4.1.1. Introduction

A **differential equation** is a mathematical equation for an unknown function of one or several variables that relates the values of the function itself and its derivatives of various orders. Differential equations are mathematically studied from several different perspectives, mostly concerned with their solutions - the set of functions that satisfy the equation. Differential equations are widely used in solving engineering problems. Many scientific laws and engineering principles and systems are in the form of or can be described by differential equations. Differential equations are mathematical tools to model engineering systems such as hydraulic flow, heat transfer, level controller of a tank, vibration isolation, electrical circuits, etc.

The linear differential equations with constant coefficients find their most important applications in the study of electrical, mechanical and other linear systems. Such equations play a prominent role in unifying the theory of electrical and mechanical oscillatory systems.

An example of modeling a real world problem using differential equations is the determination of the velocity of a ball falling through the air, considering only gravity and air resistance. The ball's acceleration towards the ground is the acceleration due to gravity minus the

deceleration due to air resistance. Gravity is considered as constant and air resistance may be modeled as proportional to the ball's velocity. This means that the acceleration of the ball is a derivative of its velocity and the velocity depends on time. Finding the velocity as a function of time involves solving a differential equation. Hence, Wherever it is possible to define a relation between a function and its derivatives, it is possible to make a mathematical model using differential equations and understand the behavior of the system related to that function.

4.1.2. Engineering Applications of Differential Equations

Exponential Growth Model - Population

Let $P(t)$ be a quantity that increases with time t and the rate of increase is proportional to the same quantity P as follows $\frac{dP}{dt} = kP$ where $\frac{dP}{dt}$ is the first derivative of P , $k > 0$ and t is the time.

The solution to the above first order differential equation is given by $P(t) = A e^{kt}$ where A is a constant not equal to 0. If $P = P_0$ at $t = 0$, then $P_0 = A e^0$ which gives $A = P_0$. The final form of the solution is given by $P(t) = P_0 e^{kt}$. Assuming P_0 is positive and since k is positive, $P(t)$ is an increasing exponential. $\frac{dP}{dt} = kP$ is also called an exponential growth model.

Exponential Decay Model - Radioactive Material

Let $M(t)$ be the amount of a product that decreases with time t and the rate of decrease is proportional to the amount M as follows $\frac{dM}{dt} = -kM$ where $\frac{dM}{dt}$ is the first derivative of M , $k > 0$ and t is the time. Solve the above first order differential equation to obtain $M(t) = A e^{-kt}$ where A is non zero constant. If we assume that $M = M_0$ at $t = 0$, then $M_0 = A e^0$ which gives $A = M_0$. The solution may be written as follows $M(t) = M_0 e^{-kt}$. Assuming M_0 is positive and since k is

positive, $M(t)$ is an decreasing exponential. $\frac{dM}{dt} = -kM$ is also called an exponential decay model.

Falling Object

An object is dropped from a height at time $t = 0$. If $h(t)$ is the height of the object at time t , $a(t)$ the acceleration and $v(t)$ the velocity. The relationships between a , v and h are as follows:

$$\frac{dv}{dt} = a(t), \quad \frac{dh}{dt} = v(t) \quad \text{for a falling object, } a(t) \text{ is constant and is equal to } g = -9.8 \text{ m/s}^2.$$

Combining the above differential equations, we can easily deduce the following equation

$$\frac{d^2h}{dt^2} = g. \quad \text{Integrate both sides of the above equation to obtain } \frac{dh}{dt} = gt + v_0. \quad \text{Integrate one more}$$

time to obtain $h(t) = \frac{gt^2}{2} + v_0t + h_0$. The above equation is a model that describes the height of a falling object, from an initial height h_0 at an initial velocity v_0 , as a function of time.

Newton's Law of Cooling

It is a model that describes mathematically, the change in temperature of an object in a given environment. The law states that the rate of change (in time) of the temperature is proportional to the difference between the temperature T of the object and the temperature T_e of the environment

$$\text{surrounding the object. } \frac{dT}{dt} = -k(T - T_e). \quad \text{Let } X = T - T_e. \quad \text{so that } \frac{dX}{dt} = \frac{dT}{dt}.$$

Using the above change of variable, the above differential equation becomes $\frac{dX}{dt} = -kX$. The

solution to the above differential equation is given by $X = Ae^{-kt}$ substitute $X = T - T_e$.

$$T - T_e = Ae^{-kt}.$$

Assume that at $t = 0$ the temperature $T = T_0$, $T_0 - T_e = A e^0$ which gives $A = T_0 - T_e$. The final expression for $T(t)$ is given by $T(t) = T_e + (T_0 - T_e)e^{-k t}$. This expression shows that how the temperature T of the object changes with time.

RL circuit

Let us consider the RL (resistor R and inductor L) circuit shown above. At time $t = 0$ the switch is closed and current passes through the circuit. Electricity laws state that the voltage across a resistor of resistance R is equal to Ri and the voltage across an inductor L is given by $L \frac{di}{dt}$ (i is the current). Another law gives an equation relating all voltages in the above circuit as follows:

$L \frac{di}{dt} + Ri = E$, where E is a constant voltage. Let us solve the above differential equation which may be written as follows.

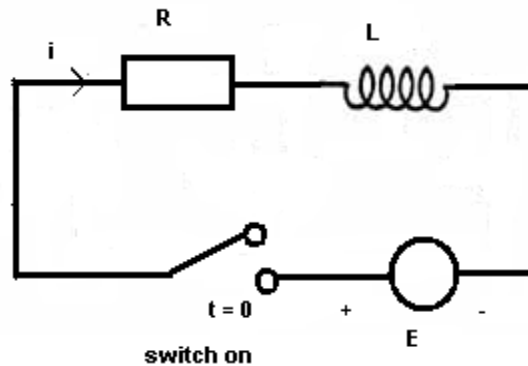


Fig. 4.1.

$\frac{L \frac{di}{dt}}{E - Ri} = 1$ which may be written as $\frac{-L}{R} \frac{d(E - Ri)}{E - Ri} = dt$. Integrate both sides

$\frac{-L}{R} \log(E - Ri) = t + c$, c constant of integration. Find constant c by setting $i = 0$ at $t = 0$ (when

switch is closed) which gives $\frac{-L}{R} \log E = c$.

Substitute c in the solution $\frac{-L}{R} \log(E - Ri) = t + \frac{-L}{R} \log E$

$$\text{(i.e.) } \frac{L}{R} \log E - \frac{L}{R} \log(E - Ri) = t$$

$$\text{(i.e.) } \log \frac{E}{E - Ri} = \frac{Rt}{L}$$

$$\text{(i.e.) } \frac{E}{E - Ri} = e^{\frac{Rt}{L}}$$

$$\text{(i.e.) } E - \frac{E}{e^{\frac{Rt}{L}}} = Ri$$

$$\text{(i.e.) } E(1 - e^{-\frac{Rt}{L}}) = Ri$$

$$\text{(i.e.) } i = \frac{E}{R}(1 - e^{-\frac{Rt}{L}})$$

The starting model for the circuit is a differential equation which when solved, gives an expression of the current in the circuit as a function of time.

Newton's second law of motion

The second law states that the net force on an object is equal to the rate of change (that is, the derivative) of its linear momentum $\mathbf{p}$ in an inertial reference frame: $F = \frac{dP}{dt} = \frac{d(mv)}{dt}$. The

second law can also be stated in terms of an object's acceleration. Since the law is valid only for constant-mass systems, the mass can be taken outside the differentiation operator by the constant

factor rule in differentiation. Thus, $F = m \frac{d(v)}{dt} = ma$, where $\mathbf{F}$ is the net force applied, m is the

mass of the body, and $\mathbf{a}$ is the body's acceleration. Thus, the net force applied to a body produces a proportional acceleration. In other words, if a body is accelerating, then there is a force on it

In this chapter, we consider the simplest of these differential equations which is of first order. We study the solutions of Exact equations, First order linear equations, Bernoulli's equations, orthogonal trajectories, growth, decay and geometrical applications. Also, we study the solutions of equations not of first degree namely, equations solvable for p , equations solvable for y , equations solvable for x and Clairaut's type.

4.1.3. Basic Definitions

Differential Equation (DE): A “differential equation” is an equation involving an unknown function y of one or more independent variables $x, t, \dots$ and its derivatives. In other words, a differential equation is an equation which involves differential co-efficients or derivatives.

Examples: 1. $\frac{dy}{dx} = \frac{1+x^2}{1+y^2}$ 2. $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 8y = 0$ 3. $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}} = k \frac{d^2y}{dx^2}$ are

differential equations.

Differential equations are classified into two categories “ordinary and partial” depending on the number of independent variables appearing in the equation.

Ordinary Differential Equations (ODE): Differential equations which involve only one independent variable and ordinary differential co-efficients are called **ordinary differential equation**.

Example: 1. $\frac{d^2y}{dx^2} + a^2x = 0$ 2. $L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{c} = E \sin \omega t$

3. $\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^3 = \left(\frac{d^2y}{dx^2} \right)^2$ are ordinary differential equations.

Partial Differential Equations (PDE): Differential equations which involve two or more independent variables and partial derivatives with respect to them are called **partial differential equations**.

Example: 1. $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$ 2. $\frac{\partial^2 z}{\partial x^2} + y \frac{\partial^2 u}{\partial y^2} = x + y$ are partial differential equations.

Order and Degree of a Differential Equation: The order of a differential equation is the order of the highest ordered derivative occurring in the differential equation. The degree of a differential equation is the degree of the highest derivative occurring in it.

For example,

1. $e^x dx + e^y dy = 0$ is of the first order first degree.

2. $\frac{d^2x}{dt^2} + n^2x = 0$ is of the second order and first degree.

3. $\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^3 = c^2 \left(\frac{d^2y}{dx^2} \right)^2$ is the second order and second degree.

4. $\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^3 = c^2 \left(\frac{d^2y}{dx^2} \right)^2$ is the second order and second degree.

5. $y \frac{dy}{dx} = x^2 \left(\frac{dy}{dx} \right)^2 + x$ is the first order and second degree.

4.2. Exact differential equation

The differential equation of a function $f(x, y)$ is denoted by df and is given by

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy \quad \dots(1)$$

Consider the differential equation

$$M(x, y)dx + N(x, y)dy = 0 \quad \dots(2)$$

Suppose there exists a function $f(x, y)$ such that

$$\frac{\partial f}{\partial x} = M(x, y) \text{ and } \frac{\partial f}{\partial y} = N(x, y) \quad \dots(3)$$

We use (3) in the given D.E (2) and it becomes

$$0 = Mdx + Ndy = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = df$$

$$(i.e.) df = 0$$

Integrating on both sides, we get $f(x, y) = c = \text{arbitrary constant.}$

In this case, the L.H.S of (2) $Mdx + Ndy$ is said to be an **exact differential** and the differential equation (2) is called an **exact differential equation**.

4.2.1. Theorem

Statement: The necessary and sufficient condition for the differential equation $Mdx + Ndy = 0$

to be exact is $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Proof: Case(i)

Necessary condition : Let $Mdx + Ndy = 0$ be an exact differential equation.

We prove that $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

By definition, the equation $Mdx + Ndy = 0$ is exact, if $du = Mdx + Ndy$ for some $u(x,y)$.

However, $du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$

$$\therefore Mdx + Ndy = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

Equating co-efficient of dx and dy , we get $M = \frac{\partial u}{\partial x}$ and $N = \frac{\partial u}{\partial y}$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial^2 u}{\partial y \partial x} \text{ and } \frac{\partial N}{\partial x} = \frac{\partial^2 u}{\partial x \partial y}$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \left(\text{Since } \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y} \right)$$

Hence proved.

Case (ii)

Sufficient condition :

Let us assume that $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$...(1)

Let $u = \int_{y \text{ constant}} M dx$

$$\therefore \frac{\partial u}{\partial x} = M \text{ and } \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial M}{\partial y}$$

$$\text{By (1), } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \text{ and } \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y}$$

$$\frac{\partial N}{\partial x} = \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right)$$

Integrating both sides with respect to x , treating y as constant, we have

$$N = \frac{\partial u}{\partial y} + f(y)$$

$$\begin{aligned} \therefore Mdx + Ndy &= \frac{\partial u}{\partial x} dx + \left[\frac{\partial u}{\partial y} + f(y) \right] dy & \left[\because M = \frac{\partial u}{\partial x}, N = \frac{\partial u}{\partial y} + f(y) \right] \\ &= \left[\frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \right] + f(y) dy \\ &= du + f(y) dy \\ &= d \left[u + \int f(y) dy \right] \end{aligned}$$

which shows that $Mdx + Ndy$ is an exact differential and hence $Mdx + Ndy = 0$ is an exact differential equation.

Note: Since $Mdx + Ndy = d \left[u + \int f(y) dy \right]$

$$Mdx + Ndy = 0 \Rightarrow d \left[u + \int f(y) dy \right] = 0$$

Integrating $u + \int f(y) dy = c$

But $u = \int Mdx$ and $f(y)$ = terms of N not containing x

Hence the solution of $Mdx + Ndy = 0$ is $\int_{y \text{ constant}} Mdx + \int (\text{terms of } N \text{ not containing } x) dy = c$

Points to Remember:-

1. A differential equation $Mdx + Ndy = 0$ is an exact differential equation if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ and is non exact differential equation if $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

2. The solution of exact differential equation is given by

$$\int_{y \text{ constant}} Mdx + \int N(\text{Terms not containing } x) dy = C$$

Example 1: Show that $(x^2 - 2xy + 3y^2)dx + (y^2 + 6xy - x^2)dy = 0$ is an exact equation and hence solve it. (Nov- 2006, Jan- 2010)

Solution:

The equation is of the form $Mdx + Ndy = 0$

$$\text{Here } M = x^2 - 2xy + 3y^2 \quad N = y^2 + 6xy - x^2$$

$$\frac{\partial M}{\partial y} = -2x + 6y \quad \frac{\partial N}{\partial x} = 6y - 2x$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Thus, the given equation is an exact equation.

$$\therefore \text{The solution is } \int_{y \text{ constant}} Mdx + \int N(\text{Terms not containing } x) dy = C$$

$$\int [x^2 - 2xy + 3y^2] dx + \int y^2 dy = C$$

$$\frac{x^3}{3} - \frac{2x^2y}{2} + 3xy^2 + \frac{y^3}{3} = C$$

$$x^3 + 9xy^2 - 3x^2y + y^3 = 3C = C_1$$

Example 2: Solve $(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2)dy = 0$

(Nov-2003, 2007, 2008, Jan-2011)

Solution:

The equation is of the form $Mdx + Ndy = 0$

Here $M = x^2 - 4xy - 2y^2$ and $N = y^2 - 4xy - 2x^2$

$$\frac{\partial M}{\partial y} = -4x - 4y \quad \frac{\partial N}{\partial x} = -4y - 4x$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Thus the given equation is an exact equation.

$\therefore$ The general solution is $\int_{y \text{ constant}} Mdx + \int N(\text{Terms not containing } x) dy = C$

$$\int (x^2 - 4xy - 2y^2)dx + \int y^2 dy = C$$

$$\frac{x^3}{3} - \frac{4x^2y}{2} - 2xy^2 + \frac{y^3}{3} = C$$

$$x^3 - 6x^2y - 6xy^2 + y^3 = C$$

Example 3: Solve $(x - 3y + 4)dx + (2y - 3x - 1)dy = 0$

(May-2011)

Solution:

The equation is of the form $Mdx + Ndy = 0$

Here $M = x - 3y + 4$ and $N = 2y - 3x - 1$

$$\frac{\partial M}{\partial y} = -3 \quad \frac{\partial N}{\partial x} = -3$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Thus the given equation is an exact equation.

$\therefore$ The general solution is $\int_{y \text{ constant}} M dx + \int N(\text{Terms not containing } x) dy = C$

$$\int (x - 3y + 4) dx + \int (2y - 1) dy = C$$

$$\frac{x^2}{2} - 3xy + 4x + \frac{2y^2}{2} - y = C$$

$$x^2 - 6xy + 8x + 2y^2 - 2y = C$$

Example 4: Solve $\frac{dy}{dx} + \frac{y \cos x + \sin y + y}{\sin x + x \cos y + x} = 0$

(Nov-2011)

Solution:

The given equation can be written as $\frac{dy}{dx} = -\frac{y \cos x + \sin y + y}{\sin x + x \cos y + x}$

$$dy(\sin x + x \cos y + x) = -dx(y \cos x + \sin y + y)$$

$$(y \cos x + \sin y + y)dx + (\sin x + x \cos y + x)dy = 0$$

The equation is of the form $Mdx + Ndy = 0$

$$M = y \cos x + \sin y + y \quad N = \sin x + x \cos y + x$$

$$\frac{\partial M}{\partial y} = \cos x + \cos y + 1 \quad \frac{\partial N}{\partial x} = \cos x + \cos y + 1$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Thus the given equation is an exact equation.

$\therefore$ The general solution is $\int_{y \text{ constant}} M dx + \int N(\text{Terms not containing } x) dy = C$

$$\int (y \cos x + \sin y + y) dx = C$$

$$y \sin x + x \sin y + yx = C$$

Example 5: Solve $(5x^4 + 3x^2y^2 - 2xy^3)dx + (2x^3y - 3x^2y^2 - 5y^4)dy = 0$

Solution:

The equation is of the form $Mdx + Ndy = 0$

Here $M = 5x^4 + 3x^2y^2 - 2xy^3$ and $N = 2x^3y - 3x^2y^2 - 5y^4$

$$\frac{\partial M}{\partial y} = 6x^2y - 6xy^2 \quad \frac{\partial N}{\partial x} = 6x^2y - 6xy^2$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Thus the given equation is an exact equation.

$\therefore$ The general solution is $\int_{y \text{ constant}} M dx + \int N(\text{Terms not containing } x) dy = C$

$$\int (5x^4 + 3x^2y^2 - 2xy^3) dx + \int -5y^4 dy = C$$

$$\frac{5x^5}{5} + \frac{3x^3y^2}{3} - \frac{2x^2y^3}{2} - \frac{5y^5}{5} = C$$

$$x^5 + x^3y^2 - x^2y^3 - y^5 = C$$

Example 6 : Solve $[\cos x \tan y + \cos(x+y)]dx + [\sin x \sec^2 y + \cos(x+y)]dy = 0$

Solution:

The equation is of the form $Mdx + Ndy = 0$

Here $M = \cos x \tan y + \cos(x + y)$ and $N = \sin x \sec^2 y + \cos(x + y)$

$$\frac{\partial M}{\partial y} = \cos x \sec^2 y - \sin(x + y) ; \quad \frac{\partial N}{\partial x} = \cos x \sec^2 y - \sin(x + y)$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Thus the given equation is an exact equation.

$\therefore$ The general solution is $\int_{y \text{ constant}} Mdx + \int N(\text{Terms not containing } x) dy = C$

$$\int [\cos x \tan y + \cos(x + y)] dx = C$$

$$\sin x \tan y + \sin(x + y) = C$$

Example 7: Solve $(x^2 - x + y^2)dx - (ye^y - 2xy)dy = 0$

(Nov-2003)

Solution:

The given equation is $(x^2 - x + y^2)dx + (2xy - ye^y)dy = 0$

The equation is of the form $Mdx + Ndy = 0$

Here $M = x^2 - x + y^2$ and $N = 2xy - ye^y$

$$\frac{\partial M}{\partial y} = 2y \qquad \frac{\partial N}{\partial x} = 2y$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Thus the given equation is an exact equation

∴ The general solution is $\int_{y \text{ constant}} Mdx + \int N(\text{Terms not containing } x)dy = C$

$$\int (x^2 - x + y^2)dx - \int ye^y dy = C$$

$$\frac{x^3}{3} - \frac{x^2}{2} + xy^2 - [ye^y - e^y] = C$$

$$\frac{x^3}{3} - \frac{x^2}{2} + xy^2 + e^y - ye^y = C$$

Exercise

Solve the following differential equations:

1. $(x^2 + y^2 - a^2)xdx + (x^2 - y^2 - b^2)ydy = 0$
2. $(3x^2 + 6xy^2)dx + (6x^2y + 4y^3)dy = 0$
3. $\left(1 + e^{\frac{x}{y}}\right)dx + \left(1 - \frac{x}{y}\right)e^{\frac{x}{y}}dy = 0$
4. $(a^2 - 2xy - y^2)dx - (x + y)^2 dx = 0$
5. $(2xy + y - \tan y)dx + (x^2 - x \tan y^2 + \sec^2 y)dy = 0$
6. $y(x^2 + y^2 + a^2)\frac{dy}{dx} + x(x^2 + y^2 - a^2) = 0$
7. $x^3 \sec^2 y \frac{dy}{dx} + 3x^2 \tan y = \cos x$

Answers

1. $x^4 + 2x^2y^2 - 2a^2x^2 - y^4 - 2b^2y^2 = C$
2. $x^3 + 3x^2y^2 + y^4 = C$
3. $x + ye^{\frac{x}{y}} = C$
4. $3a^2x - 3x^2y - 3xy^2 - y^3 = C$
5. $x^2y + xy - x \tan y + \tan y = C$
6. $x^4 + 2x^3y^2 - 2a^2x^2 + 2a^2y^3 + y^4 = C$
7. $x^3 \tan y - \sin x = C$

4.2.2. Equations Reducible to Exact Equations

Differential equations which are not exact can sometimes be made exact after multiplying by a suitable factor (a function of x and / or y) called the integrating factor (I. F).

For example, consider the equation

$$ydx - xdy = 0 \quad \dots(1)$$

Here $M = y$ and $N = -x$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} \quad \text{Therefore the equation is not an exact equation.}$$

(i) Multiplying the equation (1) by $\frac{1}{y^2}$, we get

$$\frac{ydx - xdy}{y^2} = 0 \quad (\text{or}) \quad d\left(\frac{x}{y}\right) = 0 \quad \text{which is an exact equation.}$$

(ii) Multiplying the equation (1) by $\frac{1}{x^2}$, we get $\frac{ydx - xdy}{x^2} = 0$ (or) $d\left(\frac{y}{x}\right) = 0$, which is an exact equation.

(iii) Multiplying the equation (1) by $\frac{1}{xy}$, we get $\frac{dx}{x} - \frac{dy}{y} = 0$ (or) $d(\log x - \log y) = 0$ which

is an exact equation. $\therefore \frac{1}{y^2}$, $\frac{1}{x^2}$ and $\frac{1}{xy}$ are integrating factors of (1). “If a differential

equation has one I.F., it has an infinite number of I.F.”

Important formulae:-

The following differentials are very useful in selecting a suitable integrating factor.

(i) $ydx + xdy = d(xy)$	(ii) $\frac{xdy - ydx}{x^2} = d\left(\frac{y}{x}\right)$
(iii) $\frac{ydx - xdy}{y^2} = d\left(\frac{x}{y}\right)$	(iv) $\frac{xdy - ydx}{x^2 + y^2} = d\left[\tan^{-1}\left(\frac{y}{x}\right)\right]$
(v) $\frac{xdy - ydx}{xy} = d\left[\log\left(\frac{y}{x}\right)\right]$	(vi) $\frac{ydx + xdy}{xy} = d[\log(xy)]$
(vii) $\frac{xdy + ydx}{x^2 + y^2} = d\left[\frac{1}{2}\log(x^2 + y^2)\right]$	(viii) $\frac{xdy - ydx}{x^2 - y^2} = d\left[\frac{1}{2}\log\frac{x+y}{x-y}\right]$

Example 1: Solve : $xdx = y(x^2 + y^2 - 1)dy$

(May -2009, Jan-2010)

Solution:

The given differential equation is $xdx + ydy = y(x^2 + y^2)dy$

$$\frac{2xdx + 2ydy}{x^2 + y^2} = 2ydy$$

$$\frac{d(x^2 + y^2)}{x^2 + y^2} = 2ydy$$

Integrating on both sides we get,

$$\int \frac{d(x^2 + y^2)}{x^2 + y^2} = 2 \int ydy + C$$

$$\log(x^2 + y^2) = y^2 + C$$

Example 2: Solve $(2xy^2 - y)dx + xdy = 0$

(Jan-2010)

Solution:

The given differential equation is

$$2xy^2dx + xdy - ydx = 0$$

$$2xdx + \frac{xdy - ydx}{y^2} = 0$$

$$2xdx - \left(\frac{ydx - xdy}{y^2} \right) = 0$$

$$2xdx - d\left(\frac{x}{y}\right) = 0$$

Integrating on both sides we get, $\int 2xdx - \int d\left(\frac{x}{y}\right) = C$

$$x^2 - \frac{x}{y} = C$$

$$x^2y - x = Cy$$

Example 3: Solve $xdy - ydx = x\sqrt{x^2 - y^2} dx$

Solution:

The given equation is $xdy - ydx = x^2 \sqrt{1 - \left(\frac{y}{x}\right)^2} dx$

$$\frac{xdy - ydx}{x^2} = \sqrt{1 - \left(\frac{y}{x}\right)^2} dx$$

$$d\left(\frac{y}{x}\right) = \sqrt{1 - \left(\frac{y}{x}\right)^2} dx$$

$$\frac{d\left(\frac{y}{x}\right)}{\sqrt{1 - \left(\frac{y}{x}\right)^2}} = dx$$

Integrating on both sides

$$\int \frac{d\left(\frac{y}{x}\right)}{\sqrt{1-\left(\frac{y}{x}\right)^2}} = \int dx$$

$$\sin^{-1}\left(\frac{y}{x}\right) = x + c$$

$$y = x \sin(x + c)$$

Example 4: Solve $xdx + ydy = a(x^2 + y^2)dy$

Solution:

The given equation is $\frac{xdx + ydy}{x^2 + y^2} = a dy$

$$\frac{d(x^2 + y^2)}{x^2 + y^2} = 2ady$$

Integrating on both sides,

$$\int \frac{d(x^2 + y^2)}{x^2 + y^2} = 2a \int dy + c$$

$$\log(x^2 + y^2) = 2ay + C$$

Points to remember:-

Homogeneous Equation

A function $f(x,y)$ of the variables x and y is called homogeneous of degree n , if for any parameter ' t ', $f(tx,ty) = t^n f(x,y)$.

Integrating Factor for a homogeneous equation

If $Mdx + Ndy = 0$ is a homogeneous equation in x and y , then $\frac{1}{Mx + Ny}$ is an I.F.

provided $Mdx + Ndy \neq 0$

Example 1: Solve $(x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0$

Solution:

The given equation is homogeneous in x and y

$$\text{Here } M = x^2y - 2xy^2 \text{ and } N = -x^3 + 3x^2y$$

$$\text{Now } Mx + Ny = x^3y - 2x^2y^2 - x^3y + 3x^2y^2 \neq 0$$

$$\therefore I.F. = \frac{1}{Mx + Ny} = \frac{1}{x^2y^2}$$

Multiplying throughout the given equation by $\frac{1}{(x^2y^2)}$, we get

$$\left(\frac{1}{y} - \frac{2}{x}\right)dx - \left(\frac{x}{y^2} - \frac{3}{y}\right)dy = 0 \text{ which is an exact equation}$$

$$\therefore \text{ The solution is } \int \left(\frac{1}{y} - \frac{2}{x}\right)dx + \int \frac{3}{y}dy = C$$

$$\frac{x}{y} - 2\log x + 3\log y = C$$

Example : 2 Solve $(y^3 - 2yx^2)dx + (2xy^2 - x^3)dy = 0$

Solution :

The given equation is homogeneous in x and y.

$$\text{Here } M = y^3 - 2yx^2 \text{ } N = 2xy^2 - x^3$$

$$\begin{aligned} \text{Now } Mx + Ny &= xy^3 - 2yx^3 + 2xy^3 - x^3y \\ &= 3xy^3 - 3x^3y = 3xy(y^2 - x^2) \neq 0 \end{aligned}$$

$$\therefore \text{I.F.} = \frac{1}{Mx + Ny} = \frac{1}{3xy(y^2 - x^2)}$$

Multiplying throughout the given equation by $\frac{1}{3xy(y^2 - x^2)}$, we get

$$\frac{y(y^2 - 2x^2)}{xy(y^2 - x^2)} dx + \frac{x(2y^2 - x^2)}{xy(y^2 - x^2)} dy = 0$$

$$\frac{y^2 - 2x^2}{x(y^2 - x^2)} dx + \frac{(2y^2 - x^2)}{y(y^2 - x^2)} dy = 0$$

$$\frac{(y^2 - x^2) - x^2}{x(y^2 - x^2)} dx + \frac{(y^2 - x^2) + y^2}{y(y^2 - x^2)} dy = 0$$

$$\frac{1}{x} dx - \frac{x}{y^2 - x^2} dx + \frac{1}{y} dy + \frac{y}{y^2 - x^2} dy = 0$$

$$\frac{1}{x} dx - \frac{ydy - xdx}{y^2 - x^2} + \frac{1}{y} dy = 0$$

$$\frac{1}{x} dx + \frac{1}{2} \frac{d(y^2 - x^2)}{(y^2 - x^2)} + \frac{1}{y} dy = 0$$

Multiplying throughout by 2, and integrating, we get

Points to remember:-

1.If $Mdx + Ndy = 0$ is of the form $f_1(xy)ydx + f_2(xy)x dy = 0$, then

$\frac{1}{Mx - Ny}$ is an I.F. provided $Mx - Ny \neq 0$

Example 1: Solve $y(xy + 2x^2 y^2)dx + x(xy - x^2 y^2)dy = 0$

Solution:

The given equation is of the form $f_1(xy)ydx + f_2(xy)xdy = 0$

$$\text{Here } M = xy^2 + 2x^2y^3 \quad \text{and} \quad N = x^2y - x^3y^2$$

$$\text{Now } Mx - Ny = x^2y^2 + 2x^3y^3 - x^2y^2 + x^3y^3 = 3x^3y^3 \neq 0$$

$$\therefore \text{I.F.} = \frac{1}{Mx - Ny} = \frac{1}{3x^3y^3}$$

Multiplying throughout the given equation by $\frac{1}{3x^3y^3}$, we get

$$\left(\frac{1}{3x^2y} + \frac{2}{3x} \right) dx + \left(\frac{1}{3xy^2} - \frac{1}{3y} \right) dy = 0$$

which is an exact equation.

$$\therefore \text{The solution is } \int \left(\frac{1}{3x^2y} + \frac{2}{3x} \right) dx + \int -\frac{1}{3y} dy = C$$

$$-\frac{1}{3xy} + \frac{2}{3} \log x - \frac{1}{3} \log y = C$$

$$-\frac{1}{xy} + 2 \log x - \log y = C_1, \text{ where } C_1 = 3C$$

Example 2: Solve $(1 + xy)ydx + (1 - xy)xdy = 0$

Solution:

The given equation is of the form $f_1(xy)ydx + f_2(xy)xdy = 0$

$$\text{Here } M = (1 + xy)y \quad N = (1 - xy)x$$

$$Mx - Ny = 2x^2y^2 \neq 0$$

$$\therefore \text{I.F.} = \frac{1}{Mx - Ny} = \frac{1}{2x^2y^2}$$

Multiplying throughout the given equation by $\frac{1}{2x^2y^2}$, the given equation becomes

$$\left(\frac{1}{2x^2y} + \frac{1}{2x}\right)dx + \left(\frac{1}{2xy^2} - \frac{1}{2y}\right)dy = 0 \text{ which is an exact equation.}$$

$$\therefore \text{The solution is } \int \left(\frac{1}{2x^2y} + \frac{1}{2x}\right)dx - \int \frac{1}{2y}dy = C$$

$$\frac{1}{2y} \left(-\frac{1}{x}\right) + \frac{1}{2} \log x - \frac{1}{2} \log y = C$$

$$\log \frac{x}{y} - \frac{1}{xy} = C$$

Example 3: Solve the equation $(x^3y^3 + x^2y^2 + xy + 1)ydx$

$$+ (x^3y^3 - x^2y^2 - xy + 1)x dy = 0$$

Solution:

The given equation is of the form $f_1(xy)ydx + f_2(xy)x dy = 0$

$$\text{Here } M = (x^3y^3 + x^2y^2 + xy + 1)y \quad N = (x^3y^3 - x^2y^2 - xy + 1)x$$

$$Mx - Ny = 2xy(x^2y^2 + xy) = 2x^2y^2(xy + 1) \neq 0$$

$$\therefore \text{I.F.} = \frac{1}{Mx - Ny} = \frac{1}{2x^2y^2(xy + 1)}$$

Multiplying throughout the given equation by $\frac{1}{2x^2y^2(xy + 1)}$, we get

$$\frac{(x^2y^2+1)(xy+1)}{2x^2y^2(xy+1)}ydx + \frac{(x^2y^2-1)(xy-1)}{2x^2y^2(xy+1)}xdy = 0$$

$$\left(\frac{x^2y^2+1}{2x^2y^2}\right)ydx + \left(\frac{x^2y^2-2xy+1}{2x^2y^2}\right)xdy = 0$$

$$\left(\frac{x^2y^2+1}{2x^2y^2}\right)(ydx + xdy) - \frac{1}{xy}xdy = 0$$

$$\left(\frac{1}{2} + \frac{1}{2x^2y^2}\right)d(xy) - \frac{1}{y}dy = 0$$

Integrating on both sides, we get $\frac{1}{2}xy - \frac{1}{2xy} - \log y = C$

Exercise

Solve the following Differential equations

1. $ydx - xdy + \log xdx = 0$

5. $(1+xy)ydx + (1-xy)xdy = 0$

2. $ydy + \frac{ydx - xdy}{x^2} = 0$

6. $(x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0$

3. $x dx + y dy = \frac{a^2(xdy - ydx)}{x^2 + y^2}$

7. $y(2xy + e^x)dx - e^x dy = 0$

8. $x \frac{dy}{dx} - x = x\sqrt{(x^2 + y^2)}$

4. $y(xy + 2x^2y^2)dx + x(xy - x^2y^2)dy = 0$

9. $(x^2 + y^2 + 1)dx - 2xydy = 0$

10. Show that the equation $(2x - y)dx + (2y + x)dy = 0$ can be made exact by the integrating

factor $\frac{1}{x^2 + y^2}$ and hence solve the equation

Answers

1. $x + y + \log x + 1 = 0$

$$2. \frac{y^2}{2} - \frac{y}{x} = C$$

$$3. \frac{x^2}{2} + \frac{y^2}{2} = C + a^2 \tan^{-1}\left(\frac{y}{x}\right)$$

$$4. -\frac{1}{xy} + 2 \log x - \log y = C$$

$$5. -\frac{1}{xy} + \log\left(\frac{x}{y}\right) = C$$

$$6. \frac{x}{y} - 2 \log x + 3 \log y = C$$

$$7. x^2 y + e^x = e^y$$

$$8. \log\left(\frac{y}{x} + \frac{\sqrt{x^2 + y^2}}{x}\right) = x + C$$

$$9. (x^2 - y^2 - 1) = Cx$$

$$10. \log(x^2 + y^2) + \tan^{-1}\left(\frac{y}{x}\right) = C$$

4.3. Linear Differential Equations

A differential equation is said to be linear if the dependent variable and its derivative occur only in the first degree and are not multiplied together. The general form of a linear differential equation of the first order is $\frac{dy}{dx} + Py = Q, \dots(1)$ where P and Q are functions of x only or may

be constants. Equation (1) is also known as **Leibnitz's linear equation**.

Multiplying both sides of the equation (1) by $e^{\int P dx}$, we get

$$\frac{dy}{dx} e^{\int P dx} + y(e^{\int P dx} P) = Q e^{\int P dx} \quad \text{or}$$

$$\frac{d}{dx}(y e^{\int P dx}) = Q e^{\int P dx}$$

Integrating both sides, we get $y e^{\int P dx} = \int Q e^{\int P dx} dx + c$, which is the required solution.

Note 1: The factor $e^{\int P dx}$ is called the integrating factor.

Thus I.F. = $e^{\int p dx}$ and the solution is $y(I.F) = \int Q(I.F) dx + c$

Note 2: Sometimes a differential equation takes linear form if we regard x as dependent variable and y as independent variable. Then the equation can be put as $\frac{dx}{dy} + Px = Q$, where P, Q are functions of y only or constants.

The integrating factor in this case is $e^{\int p dy}$ and the solution is $x(I.F) = \int Q(I.F) dy + c$

Note 3: In the general form of a linear differential equation, the co-efficient of $\frac{dy}{dx}$ is unity. For example, the equation $R \frac{dy}{dx} + Sy = T$, where R, S and T are functions of x only or constants, must be divided by R to bring it to the general linear form.

Points to remember:-

1. The general form of a linear differential equation of the first order is $\frac{dy}{dx} + Py = Q$ and its solution is given by $ye^{\int P dx} = \int Qe^{\int P dx} dx + c$,

Example 1: Solve $\frac{1}{x} \frac{dy}{dx} + \frac{y}{x} \tan x = \cos x$

(Jan- 2010)

Solution:

Multiplying on both sides by x to make the co-efficient of $\frac{dy}{dx}$ unity, we get

$$\frac{dy}{dx} + y \tan x = x \cos x$$

Comparing it with $\frac{dy}{dx} + Py = Q$, we have

$$P = \tan x, \quad Q = x \cos x$$

$$\int p dx = \int \tan x dx = \log(\sec x)$$

$$e^{\int p dx} = e^{\log(\sec x)} = \sec x$$

$$\therefore \text{The solution is } ye^{\int p dx} = \int Qe^{\int p dx} dx + C$$

$$y \sec x = \int x \cos x \sec x dx + C$$

$$= \int x dx + C$$

$$y \sec x = \frac{x^2}{2} + C$$

Example 2: Solve $(1 + y^2)dx = (\tan^{-1} y - x)dy$

(Nov-2005, 2006, May- 2008)

Solution:

The given equation can be written as $(1 + y^2) \frac{dx}{dy} + x = \tan^{-1} y$

Dividing by $(1 + y^2)$ we get $\frac{dx}{dy} + \frac{x}{1 + y^2} = \frac{\tan^{-1} y}{1 + y^2}$

This is of the form $\frac{dx}{dy} + px = Q$

$$\text{Here } P = \frac{1}{1 + y^2}, \quad Q = \frac{\tan^{-1} y}{1 + y^2}$$

$$\text{I.F.} = e^{\int p dy} = e^{\int \frac{1}{1+y^2} dy} = e^{\tan^{-1} y}$$

$\therefore$ The solution is $xe^{\int p dy} = \int Qe^{\int p dy} dy + C$

$$xe^{\tan^{-1} y} = \int \frac{\tan^{-1} y}{1+y^2} e^{\tan^{-1} y} dy + C$$

Let $t = \tan^{-1} y$

$$dt = \frac{1}{1+y^2} dy$$

$$\therefore xe^{\tan^{-1} y} = \int te^t dt + C$$

$$= te^t - \int 1 \cdot e^t dt + C$$

$$= te^t - e^t + C = (\tan^{-1} y - 1)e^{\tan^{-1} y} + C$$

$$\therefore x = \tan^{-1} y - 1 + Ce^{-\tan^{-1} y}$$

Example 3: Solve $x(1-x^2)\frac{dy}{dx} + (2x^2-1)y = x^3$

(Apr- 2010)

Solution:

Given that $x(1-x^2)\frac{dy}{dx} + (2x^2-1)y = x^3$

Dividing by $x(1-x^2)$, we get

$$\frac{dy}{dx} + \frac{2x^2-1}{x(1-x^2)} y = \frac{x^2}{1-x^2}$$

This is of the form $\frac{dy}{dx} + Py = Q$

$$\text{Here } P = \frac{2x^2-1}{x(1-x)(1+x)} = \frac{-1}{x} + \frac{1}{2(1-x)} + \frac{-1}{2(1+x)}$$

$$\begin{aligned}
\int p dx &= -\log x - \frac{1}{2} \log(1-x) - \frac{1}{2} \log(1+x) \\
&= -\log \left[x(1-x)^{1/2}(1+x)^{1/2} \right] \\
&= -\log \left[x(1-x^2)^{1/2} \right] = \log \frac{1}{x\sqrt{1-x^2}}
\end{aligned}$$

$$\begin{aligned}
\text{I.F.} &= e^{\int p dx} = e^{\log \frac{1}{x\sqrt{1-x^2}}} \\
&= \frac{1}{x\sqrt{1-x^2}}
\end{aligned}$$

$\therefore$ The solution is $ye^{\int p dx} = \int Qe^{\int p dx} dx + C$

$$\begin{aligned}
y \cdot \frac{1}{x\sqrt{1-x^2}} &= \int \frac{x^2}{1-x^2} \times \frac{1}{x\sqrt{1-x^2}} dx + C \\
&= -\frac{1}{2} \int (1-x^2)^{-3/2} (-2x) dx + C \\
&= (1-x^2)^{-1/2} + C \\
y &= x + Cx\sqrt{1-x^2}
\end{aligned}$$

Example 4: Solve $\left(\frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}} \right) \frac{dx}{dy} = 1$

Solution:

The given equation can be written as $\frac{dy}{dx} + \frac{y}{\sqrt{x}} = \frac{e^{-2\sqrt{x}}}{\sqrt{x}}$

This is of the form $\frac{dy}{dx} + Py = Q$

$$\text{Here } P = \frac{1}{\sqrt{x}}, \quad Q = \frac{e^{-2\sqrt{x}}}{\sqrt{x}}$$

$$\text{I.F.} = e^{\int P dx} = e^{\int x^{-1/2} dx} = e^{2\sqrt{x}}$$

$$\therefore \text{The solution is } ye^{\int P dx} = \int Qe^{\int P dx} dx + C$$

$$ye^{2\sqrt{x}} = \int \frac{e^{-2\sqrt{x}}}{\sqrt{x}} e^{2\sqrt{x}} dx + C$$

$$\begin{aligned} ye^{2\sqrt{x}} &= \int x^{-1/2} dx + C \\ &= 2\sqrt{x} + C \end{aligned}$$

Example 5: Solve $\frac{dy}{dx} + \frac{y}{x} = xe^x$

(May- 2001, 2004)

Solution:

The given equation is of the form $\frac{dy}{dx} + Py = Q$

$$\text{Here } P = \frac{1}{x}, \quad Q = xe^x$$

$$\therefore \text{I.F.} = e^{\int P dx} = e^{\int \frac{1}{x} dx} = e^{\log x} = x$$

$$\therefore \text{The solution is } ye^{\int P dx} = \int Qe^{\int P dx} dx + C$$

$$yx = \int (xe^x) x dx + C$$

$$xy = \int x^2 e^x dx + C$$

$$\text{Let } u = x^2, \quad dv = e^x dx$$

$$u' = 2x, \quad v = e^x$$

$$u'' = 2, \quad v_1 = e^x$$

$$v_2 = e^x$$

$$xy = [uv - u'v_1 + u''v_2] + C$$

$$= [x^2 e^x - 2xe^x + 2e^x] + C$$

$$xy = (x^2 - 2x + 2)e^x + C$$

Example 6: Solve $\cos^2 x \frac{dy}{dx} + y = \tan x$

(May- 2002)

Solution:

Dividing by $\cos^2 x$, we get

$$\frac{dy}{dx} + \frac{1}{\cos^2 x} y = \frac{\tan x}{\cos^2 x}$$

$$\frac{dy}{dx} + (\sec^2 x) y = \frac{\tan x}{\cos^2 x}$$

$$\frac{dy}{dx} + (\sec^2 x) y = \tan x \sec^2 x$$

This is of the form $\frac{dy}{dx} + Py = Q$

$$\text{Here } P = \sec^2 x \quad Q = \tan x \sec^2 x$$

$$\text{I.F.} = e^{\int P dx} = e^{\int \sec^2 x dx} = e^{\tan x}$$

$\therefore$ The solution is $ye^{\int p dx} = \int Qe^{\int p dx} dx + C$

$$ye^{\tan x} = \int \tan x \sec^2 x e^{\tan x} dx + C$$

$$\text{Let } t = \tan x; \quad dt = \sec^2 x dx$$

$$ye^{\tan x} = \int t e^t dt + C$$

$$\text{Put } u = t \quad dv = e^t dt$$

$$u' = 1 \quad v = e^t$$

$$v_1 = e^t$$

$$ye^{\tan x} = [uv - u'v_1] + C$$

$$= [te^t - e^t] + C$$

$$= (t-1)e^t + C$$

$$ye^{\tan x} = [\tan x - 1]e^{\tan x} + C$$

Example 7: Solve $\sec x \frac{dy}{dx} = y + \sin x$

(May -2011)

Solution:

The given equation can be written as $\sec x \frac{dy}{dx} - y = \sin x$

Multiplying by $\cos x$, we get

$$\frac{dy}{dx} - y \cos x = \sin x \cos x,$$

This is of the form $\frac{dy}{dx} + Py = Q$

Here $P = -\cos x$, $Q = \sin x \cos x$

$$\text{I.F.} = e^{\int P dx} = e^{-\int \cos x dx} = e^{-\sin x}$$

$\therefore$ The solution is $ye^{\int P dx} = \int Qe^{\int P dx} dx + C$

$$ye^{-\sin x} = \int \sin x \cos x e^{-\sin x} dx + C$$

Let $t = \sin x$ $dt = \cos x dx$

$$ye^{-t} = \int te^{-t} dt + C$$

$$= -te^{-t} - e^{-t} + C$$

$$= e^{-t}(-t-1) + C$$

$$\therefore ye^{-\sin x} = e^{-\sin x}(-\sin x - 1) + C$$

Exercise

Solve the following differential equations:

1. $\frac{dy}{dx} + \frac{y}{x} = x^3 - 3$

if $y = 4$ when $x = 0$

2. $\frac{dy}{dx} + y \cot x = \cos x$

6. $(1+x^3)\frac{dy}{dx} + 6x^2y = 1+x^2$

3. $x\frac{dy}{dx} + y = e^x - xy$

7. $(x^2+1)\frac{dy}{dx} + 2xy = x^2$

4. $(x+1)\frac{dy}{dx} - y = e^x(x+1)^2$

8. $x \log x \frac{dy}{dx} + y = 2 \log x$

5. $\frac{dy}{dx} - y \tan x = 3e^{-\sin x}$

Answers

1. $10xy = 2x^5 - 15x^2 + C$

5. $y = \tan x - 1 + Ce^{-\tan x}$

2. $y \sin x = \frac{1}{2} \sin^2 x + C$

6. $y(1+x^3)^2 = x + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^6}{6} + C$

3. $xy = \frac{1}{2} e^x + Ce^{-x}$

7. $y(x^2 + 1) = \frac{x^3}{3} + C$

4. $y = (x+1)e^x + C$

8. $y \log x = (\log x)^2 + C$

4.4. Nonlinear Equation Reducible to the Linear form**4.4.1. Bernoulli's Equation**

Consider the equation $\frac{dy}{dx} + Py = Qy^n \dots(1)$ where P and Q are functions of x alone or constants. We will show that this equation can be made linear by changing the dependent variable.

Dividing (1) throughout by y^n , we get $\frac{1}{y^n} \frac{dy}{dx} + \frac{P}{y^n} y = Q$

$$\text{(i.e.) } y^{-n} \frac{dy}{dx} + Py^{1-n} = Q \quad \dots(2)$$

$$\text{Let } y^{1-n} = v \text{ where } v \text{ is a new variable} \quad \dots(3)$$

Differentiating (3) with respect to x,

$$(1-n)y^{1-n-1} \frac{dy}{dx} = \frac{dv}{dx}$$

$$\text{(i.e.) } y^{-n} \frac{dy}{dx} = \frac{1}{1-n} \frac{dv}{dx} \quad \dots(4)$$

Substituting (3) and (4) in (2), we get

$$\frac{1}{1-n} \frac{dv}{dx} + Pv = Q$$

$$\text{(i.e.) } \frac{dv}{dx} + P(1-n)v = Q(1-n) \quad \dots(5)$$

Now (5) is clearly a linear equation in v and the integrating factors is $e^{\int P(1-n)dx}$

Multiplying both sides of (5) by $e^{\int P(1-n)dx}$ we have $\frac{d}{dx} \left[v e^{\int P(1-n)dx} \right] = Q(1-n) e^{\int P(1-n)dx}$

On integration, we get $v e^{\int P(1-n)dx} = \int Q(1-n) e^{\int P(1-n)dx} dx + C$

(i.e.) $y^{1-n} e^{\int P(1-n)dx} = \int Q(1-n) e^{\int P(1-n)dx} dx + C$. This forms the solution of equation (1).

Note: Equations of the form $f'(y) \frac{dy}{dx} + Pf(y) = Q$, where P and Q are functions of x alone or

constants, can also be reduced to the linear type by changing the dependent variable in the

following manner. Putting $f(y) = v$, we have $f'(y) \frac{dy}{dx} = \frac{dv}{dx}$

The above equation becomes $\frac{dv}{dx} + Pv = Q$ which is linear in v

Example 1: Solve $x \frac{dy}{dx} + y = x^3 y^6$

(Nov-2002, 2008, Jan- 2011)

Solution:

Given that $x \frac{dy}{dx} + y = x^3 y^6$

Dividing by x , we get $\frac{dy}{dx} + \frac{1}{x}y = x^2y^6$... (1)

Dividing (1) by y^6 , we have

$$y^{-6} \frac{dy}{dx} + \frac{1}{x} y^{-5} = x^2 . \quad \dots (2)$$

Put $v = y^{-5}$

$$\frac{dv}{dx} = -5y^{-6} \frac{dy}{dx}$$

$$y^{-6} \frac{dy}{dx} = -\frac{1}{5} \frac{dv}{dx}$$

$\therefore$ (2) becomes $-\frac{1}{5} \frac{dv}{dx} + \frac{1}{x}v = x^2$

(i.e.) $\frac{dv}{dx} - \frac{5}{x}v = -5x^2$. This is linear equation in v .

Hence $P = \frac{-5}{x}$, $Q = -5x^2$

$$\int P dx = \int \frac{-5}{x} dx = -5 \log x = \log x^{-5}$$

$$\text{I.F.} = e^{\int P dx} = e^{\log x^{-5}} = x^{-5} = \frac{1}{x^5}$$

$\therefore$ The solution is $ve^{\int P dx} = \int Qe^{\int P dx} dx + C$

$$\frac{y^{-5}}{x^5} = \int (-5x^2) \frac{1}{x^5} dx + C$$

$$\frac{1}{x^5 y^5} = -5 \left[\frac{x^{-2}}{-2} \right] + C$$

$$\frac{1}{x^5 y^5} = \frac{5}{2x^2} + C$$

Example 2: Solve $2y \cos y^2 \frac{dy}{dx} - \frac{2}{x+1} \sin y^2 = (x+1)^3$

Solution:

Put $v = \sin y^2$

$$\frac{dv}{dx} = \cos y^2 2y \frac{dy}{dx}$$

$\therefore$ The given equation becomes $\frac{dv}{dx} - \frac{2}{x+1} v = (x+1)^3$. This is linear equation in v.

$$\text{Here } P = \frac{-2}{x+1}, \quad Q = (x+1)^3$$

$$\int P dx = -2 \log(x+1) = \log \frac{1}{(x+1)^2}$$

$$\therefore \text{ I.F.} = e^{\int P dx} = e^{\log \frac{1}{(x+1)^2}} = \frac{1}{(x+1)^2}$$

$$\therefore \text{ The solution is } v e^{\int P dx} = \int Q e^{\int P dx} dx + C$$

$$v \frac{1}{(x+1)^2} = \int (x+1)^3 \frac{1}{(x+1)^2} dx + C = \int (x+1) dx + C$$

$$\sin y^2 \frac{1}{(x+1)^2} = \frac{x^2}{2} + x + C$$

Example 3: Solve $\frac{dy}{dx} + \frac{x}{1-x^2} y = x\sqrt{y}$

(Nov- 2003)

Solution:

Dividing by $\sqrt{y}$, the given equation becomes $y^{-1/2} \frac{dy}{dx} + \frac{x}{1-x^2} y^{1/2} = x$... (1)

Put $v = y^{1/2}$

$$\frac{dv}{dx} = \frac{1}{2} y^{-1/2} \frac{dy}{dx}$$

$$y^{-1/2} \frac{dy}{dx} = 2 \frac{dv}{dx}$$

$\therefore$ (1) becomes $2 \frac{dv}{dx} + \frac{x}{1-x^2} v = x$

$\frac{dv}{dx} + \frac{x}{2(1-x^2)} v = \frac{x}{2}$. This is linear equation in v

Here $P = \frac{x}{2(1-x^2)}$, $Q = \frac{x}{2}$

$$\int P dx = \int \frac{x}{2(1-x^2)} dx = \frac{1}{-4} \int \frac{-2x}{1-x^2} dx = -\frac{1}{4} \log(1-x^2)$$

$$= \log(1-x^2)^{-\frac{1}{4}}$$

$$\therefore \text{I.F.} = e^{\int P dx} = (1-x^2)^{-\frac{1}{4}}$$

$\therefore$ The solution is $ve^{\int P dx} = \int Qe^{\int P dx} dx + C$

$$\begin{aligned}
 v(1-x^2)^{-\frac{1}{4}} &= \int \frac{x}{2}(1-x^2)^{-\frac{1}{4}} dx + C \\
 &= -\frac{1}{4} \int (-2x)(1-x^2)^{-\frac{1}{4}} dx + C
 \end{aligned}$$

put $1-x^2 = t$, $-2x dx = dt$

$$= -\frac{1}{4} \int t^{-\frac{1}{4}} dt + C = -\frac{1}{4} \left(\frac{t^{\frac{3}{4}}}{\frac{3}{4}} \right) + C$$

$$v(1-x^2)^{-\frac{1}{4}} = -\frac{1}{4} \frac{(1-x^2)^{\frac{3}{4}}}{\frac{3}{4}} + C$$

$$v = -\frac{1}{3}(1-x^2) + C(1-x^2)^{\frac{1}{4}}$$

$$\therefore \sqrt{y} = -\frac{1}{3}(1-x^2) + C(1-x^2)^{\frac{1}{4}}$$

Example 4: Solve $\frac{dy}{dx} - y \tan x = \frac{\sin x \cos^2 x}{y^2}$

(Nov-2004, May -2007)

Solution:

Multiplying by y^2 , the given equation becomes

$$y^2 \frac{dy}{dx} - y^3 \tan x = \sin x \cos^2 x \quad \dots(1)$$

Put $v = y^3$

$$\frac{dv}{dx} = 3y^2 \frac{dy}{dx}$$

$$y^2 \frac{dy}{dx} = \frac{1}{3} \frac{dv}{dx}$$

Equation (1) becomes $\frac{1}{3} \frac{dv}{dx} - v \tan x = \sin x \cos^2 x$

(i.e.) $\frac{dv}{dx} - (3 \tan x)v = 3 \sin x \cos^2 x$. This is linear equation in v

Here $P = -3 \tan x$, $Q = 3 \sin x \cos^2 x$

$$\int P dx = -3 \int \tan x dx = -3 \log(\sec x) = \log(\sec^{-3} x)$$

$$e^{\int P dx} = \sec^{-3} x = \cos^3 x$$

$\therefore$ The solution is $ve^{\int P dx} = \int Qe^{\int P dx} dx + C$

$$y^3 \cos^3 x = \int 3 \sin x \cos^2 x (\cos^3 x) du + C$$

$$= 3 \int \cos^5 x \sin x dx + C$$

Let $t = \cos x$, $dt = -\sin x dx$

$$\therefore y^3 \cos^3 x = 3 \int t^5 (-dt) + C$$

$$= -\frac{3t^6}{6} + C$$

$$\therefore y^3 \cos^3 x = -\frac{\cos^6 x}{2} + C$$

Example 5: Solve $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$

Solution:

Dividing by $\cos^2 y$, the given equation becomes

$$\sec^2 x \frac{dy}{dx} + x \frac{2 \sin y \cos y}{\cos^2 y} = x^3$$

$$\sec^2 x \frac{dy}{dx} + 2x \tan y = x^3 \quad \dots(1)$$

Put $v = \tan y$

$$\frac{dv}{dx} = \sec^2 y \frac{dy}{dx}$$

Equation (1) becomes $\frac{dv}{dx} + 2xv = x^3$. This is linear equation in v . Here $P = 2x$, $Q = x^3$

$$\text{I.F.} = e^{\int 2x dx} = e^{x^2}$$

$\therefore$ The solution is $ve^{\int P dx} = \int Qe^{\int P dx} dx + C$

$$ve^{x^2} = \int x^3 e^{x^2} dx + C = \int x^2 e^{x^2} x dx + C$$

$$\text{put } x^2 = t, 2x dx = dt$$

$$= \frac{1}{2} \int t e^t dt + C$$

$$= \frac{1}{2} (t - 1) e^t + C$$

$$ve^{x^2} = \frac{1}{2} (x^2 - 1) e^{x^2} + C$$

$$\therefore \tan y e^{x^2} = \frac{1}{2} (x^2 - 1) e^{x^2} + C$$

$$\tan y = \frac{1}{2} (x^2 - 1) + C e^{-x^2}$$

Example 6: Solve $\frac{dy}{dx} + \frac{y}{x} = xe^x$

Solution:

This is linear equation in y

$$\text{Here } P = \frac{1}{x}, \quad Q = xe^x$$

$$e^{\int P dx} = e^{\int \frac{1}{x} dx} = e^{\log x} = x$$

$$\therefore \text{ The solution is } ye^{\int P dx} = \int Qe^{\int P dx} dx + C$$

$$yx = \int xe^x x dx + C$$

$$= \int x^2 e^x dx + C$$

$$\text{Put } u = x^2 \quad v = e^x$$

$$u' = 2x \quad v_1 = e^x$$

$$u'' = 2 \quad v_2 = e^x$$

$$\therefore yx = [uv - u'v_1 + u''v_2] + C$$

$$xy = [x^2 e^x - 2xe^x + 2e^x] + C$$

$$xy = [x^2 - 2x + 2]e^x + C$$

Example 7: Solve $\frac{dy}{dx} - 2y \tan x = y^2 \cos^3 x$

Solution:

Dividing by y^2 , we get

$$\frac{1}{y^2} \frac{dy}{dx} - \frac{2 \tan x}{y} = \cos^3 x$$

$$y^{-2} \frac{dy}{dx} - 2 \tan x y^{-1} = \cos^3 x$$

$$\text{Put } v = y^{-1}$$

$$\frac{dv}{dx} = -y^{-2} \frac{dy}{dx}$$

$$y^{-2} \frac{dy}{dx} = -\frac{dv}{dx}$$

$$\therefore (1) \text{ becomes } -\frac{dv}{dx} - 2 \tan x v = \cos^3 x$$

$$\frac{dv}{dx} + 2 \tan x v = -\cos^3 x. \text{ This is linear equation in } v.$$

$$\text{Here } P = 2 \tan x, \quad Q = -\cos^3 x$$

$$\int P dx = 2 \int \tan x dx = 2 \log(\sec x) = \log(\sec^2 x)$$

$$\therefore \text{ I.F. } = e^{\int P dx} = \sec^2 x$$

$$\therefore \text{ The solution is } v e^{\int P dx} = \int Q e^{\int P dx} dx + C$$

$$y^{-1} \sec^2 x = \int -\cos^3 x \sec^2 x dx + C$$

$$= -\int \cos x dx + C$$

$$\frac{\sec^2 x}{y} = -\sin x + C$$

Example 8: Solve $xy - \frac{dy}{dx} = y^3 e^{-x^2}$

Solution:

The given equation can be written as $\frac{dy}{dx} - xy = -y^3 e^{-x^2}$

Dividing by y^3 , we get $\frac{1}{y^3} \frac{dy}{dx} - \frac{x}{y^2} = -e^{-x^2}$

$$y^{-3} \frac{dy}{dx} - xy^{-2} = -e^{-x^2} \quad \dots(1)$$

Let $v = y^{-2}$

$$\frac{dv}{dx} = -2y^{-3} \frac{dy}{dx}$$

$$y^{-3} \frac{dy}{dx} = -\frac{1}{2} \frac{dv}{dx}$$

Equation (1) becomes $-\frac{1}{2} \frac{dv}{dx} - xv = -e^{-x^2}$

$\frac{dv}{dx} + 2xv = 2e^{-x^2}$. This is linear equation in v

Here $P = 2x$, $Q = 2e^{-x^2}$

$$\text{I.F.} = e^{\int P dx} = e^{\int 2x dx} = e^{x^2}$$

$\therefore$ The solution is $ve^{\int P dx} = \int Qe^{\int P dx} dx + C$

$$y^{-2} e^{x^2} = \int 2e^{-x^2} e^{x^2} dx + C$$

$$\frac{e^{x^2}}{y^2} = 2x + C$$

Example 9: Solve $\frac{1}{y^3} \left(\frac{dy}{dx} \right) + \frac{1}{y^2} \left(\frac{2}{x} \right) = \frac{1}{x^3}$ **(Jan-2013)**

Solution:

The given equation can be written as $y^{-3} \left(\frac{dy}{dx} \right) + y^{-2} \left(\frac{2}{x} \right) = \frac{1}{x^3}$... (1)

$$\text{Put } v = y^{-2}$$

$$\frac{dv}{dx} = -2y^{-3} \frac{dy}{dx}$$

$$y^{-3} \frac{dy}{dx} = -\frac{1}{2} \frac{dv}{dx}$$

$$\text{Equation (1) becomes } -\frac{1}{2} \frac{dv}{dx} + \frac{2}{x} v = \frac{1}{x^3}$$

$$\text{(i.e.) } \frac{dv}{dx} - \frac{4}{x} v = -\frac{2}{x^3} . \text{ This is a linear equation in v.}$$

$$\text{Here } P = -\frac{4}{x}, \quad Q = -\frac{2}{x^3}$$

$$\text{I.F.} = e^{\int P dx} = e^{-4 \int \frac{1}{x} dx} = e^{-4 \log x} = e^{\log x^{-4}} = \frac{1}{x^4}$$

$$\therefore \text{ The solution is } v e^{\int P dx} = \int Q e^{\int P dx} dx + C$$

$$y^{-2} \frac{1}{x^4} = \int -\frac{2}{x^3} \frac{1}{x^4} dx + C$$

$$\frac{1}{x^4 y^2} = -2 \int \frac{1}{x^7} dx + C$$

$$\frac{1}{x^4 y^2} = -\frac{1}{3x^6} + C$$

Exercise

Solve the following equations

1. $(1-x^2) \frac{dy}{dx} - xy = x^2 y^2$

4. $2y^2 \frac{dy}{dx} + \frac{y^4}{x} = e^x$

2. $\frac{dy}{dx} + y \cos x = y^n \sin 2x$

5. $3 \frac{dy}{dy} + y = e^{3x} y^4$

3. $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$

6. $\frac{dy}{dx} + y \tan x = y^2 \sin^2 x$

Answers

1. $\frac{1}{y} = (C + \sin^{-1} x) \sqrt{1-x^2} - x$

4. $y^4 x^2 = 2e^x (x^2 - 2x + 2) + C$

2.

5. $y^{-3} = Ce^x - \frac{1}{2} e^{3x}$

$y^{1-n} e^{(1-n) \sin x} = \frac{2e^{(1-n) \sin x}}{1-n} [(1-n) \sin x - 1] + C$

6. $\frac{\cos x}{y} = \frac{C - \sin^3 x}{3}$

3. $\tan e^{x^2} = \frac{1}{2} e^{x^2} (x^2 - 1) + C$

4.5. Orthogonal Trajectories

4.5.1. Orthogonal Trajectories in Cartesian Co-ordinates

If two families of curves are such that each member of either family cuts each member of the other family at right angles, the members of one family are known as the **Orthogonal**

Trajectories of the other. Families of curves related in this way occur frequently in applied problems. For instance, in two dimensional problems in the flow of heat, the curves along which the heat flow takes place and the isothermal curves or loci of points at the same temperature are orthogonal trajectories. Similarly, in problems in the flow of electricity in thin conducting sheets, the paths along which the current flows are orthogonal trajectories of the equipotential curves and vice versa. In hydrodynamics, the flow of water from a lake into a narrow channel produces a family of streamlines which are the orthogonal trajectories to the curves of equal velocity potential. Consider $f(x, y, c) = 0$... (1)

represents one family of curves, c being an arbitrary constant. We can form a first order differential equation which is independent of c , and whose general solution is (1)

Let this differential equation be $F\left(x, y, \frac{dy}{dx}\right) = 0$... (2)

We know that if two curves intersect at right angles, the product of their slopes must be equal to -1 . Now let (x, y) be the point of intersection of a curve represented by (1), and its orthogonal trajectory. The values of x and y are the same for both the curves. $\frac{dy}{dx}$ is the slope of

the first curve and $-\frac{1}{\left(\frac{dy}{dx}\right)}$ will be the slope of the orthogonal trajectory.

Therefore, if we write $-\frac{1}{\left(\frac{dy}{dx}\right)}$ for $\frac{dy}{dx}$ in (2) keeping x and y the same, we will get the

differential equation of the family of orthogonal trajectories of the family represented by (1). Hence we get the differential equations of the system of orthogonal trajectories as

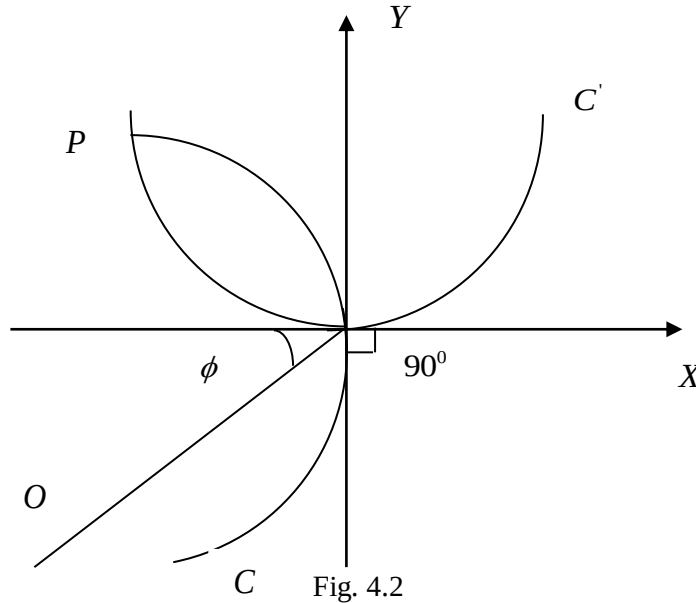
$$F\left[x, y, -\frac{1}{\left(\frac{dy}{dx}\right)}\right] = 0 \quad \text{or} \quad F\left[x, y, -\frac{dx}{dy}\right] = 0 \quad \dots(3)$$

Thus the problem of finding the O.T. of a given family of curves requires the solution of a first order differential equation.

4.5.2. Orthogonal Trajectories – Polar co-ordinates

$$\text{Let } f(r, \theta, k) = 0 \quad \dots(4)$$

represent one family of curves in polar Co-ordinates and C be a particular member of this family. By differentiating the equation (4) with respect to θ , we can eliminate the parameter k and obtain the differential equation of the given family.



$$\text{Let this differential equation be } F\left(r, \theta, \frac{dr}{d\theta}\right) = 0 \quad \dots(5)$$

Now let C' be a curve which cuts orthogonally the curve C at P . If ϕ is the angle between the tangent at P to C and the radius vector OP , we know that $\tan \phi = \frac{r d\theta}{dr}$. If ϕ' is the angle between the tangent at P to the curve C' and the radius OP , we have $\phi' = \phi + 90^\circ$ as the two tangents are at right angles.

Here $\tan \phi' = \tan(90^\circ + \phi) = -\cot \phi$

$$= -\frac{1}{\tan \phi} = -\frac{1}{r \frac{d\theta}{dr}} = -\frac{1}{r} \frac{dr}{d\theta}$$

Hence, if we write $-\frac{1}{r} \frac{dr}{d\theta}$ in the place of $r \frac{d\theta}{dr}$ in equation (5), keeping r and θ the same, we will get the differential equation of the family of orthogonal trajectories of the family represented by (4). Obviously we replace $\frac{dr}{d\theta}$ in the equation (5) by $-r^2 \frac{d\theta}{dr}$. We get the differential equation of the system of orthogonal trajectories as $F\left(r, \theta, -r^2 \frac{d\theta}{dr}\right) = 0$.

Note: A family of curves $f(x_1, y_1, c_1) = 0$ is said to be self orthogonal, if the same differential equation is obtained by replacing $\frac{dy}{dx} = -\frac{dx}{dy}$ in the differential equation of the form $f(x_2, y_2, c_2) = 0$

Points to remember:-

1. Orthogonal trajectory in Cartesian form

The differential equations of the system of orthogonal trajectories is given by

$$F\left[x, y, -\frac{1}{\left(\frac{dy}{dx}\right)}\right] = 0 \quad \text{or} \quad F\left[x, y, -\frac{dx}{dy}\right] = 0$$

2. Orthogonal trajectory in Polar form

The differential equations of the system of orthogonal trajectories is given by

$$F\left(r, \theta, -r^2 \frac{d\theta}{dr}\right) = 0$$

Example 1: Find the orthogonal trajectories of family of rectangular hyperbola $xy = c^2$

Solution:

The given equation is $xy = c^2$... (1)

Differentiating with respect to x,

$$x \frac{dy}{dx} + y = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$ we get

$$-\frac{dx}{dy} = -\frac{y}{x}$$

$$\frac{dy}{dx} = \frac{x}{y}$$

$$ydy = xdx$$

Integrating, $\int ydy = \int xdx$

$$\frac{y^2}{2} = \frac{x^2}{2} + C$$

$y^2 - x^2 = 2C$ which is the required orthogonal trajectories.

Example 2: Find the orthogonal trajectories of family of circle $x^2 + y^2 = a^2$

(May-2002)

Solution :

The given equation is $x^2 + y^2 = a^2$

Differentiating with respect to 'x',

$$2x + 2y \frac{dy}{dx} = 0$$

$$2y \frac{dy}{dx} = -2x$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$, we get

$$-\frac{dx}{dy} = -\frac{x}{y}$$

$$\frac{dy}{dx} = +\frac{y}{x}$$

$$\frac{dy}{y} = \frac{dx}{x}$$

Integrating on both sides, we get $\int \frac{dy}{y} = \int \frac{dx}{x}$

$$\log y = \log x + \log c$$

$$\log y - \log x = \log c$$

$$\frac{y}{x} = C \text{ which is the required orthogonal trajectories.}$$

Example 3: Find the orthogonal trajectories of the system of Parabolas $y^2 = 4ax$

(Apr-2003, Nov-2003, 2006)

Solution:

The given equation is $y^2 = 4ax$

Differentiating with respect to x,

$$2y \frac{dy}{dx} = 4a$$

$$\frac{dy}{dx} = \frac{2a}{y}$$

$$\frac{dy}{dx} = \frac{2y^2}{4xy} = \frac{y}{2x} \quad \left[\because a = \frac{y^2}{4x} \right]$$

Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$, we get

$$-\frac{dx}{dy} = \frac{y}{2x}$$

$$\frac{dy}{dx} = -\frac{2x}{y}$$

Integrating on both sides, we get $\int y dy = -2 \int x dx$

$$\frac{y^2}{2} = -\frac{2x^2}{2} + C \text{ which is the required orthogonal trajectories.}$$

Example 4: Find the O.T. of the family of ellipses given by $\frac{x^2}{a^2} + \frac{y^2}{b^2 + \lambda} = 1$

Where λ is a parameter

(May-2002)

Solution:

The given equation is $\frac{x^2}{a^2} + \frac{y^2}{b^2 + \lambda} = 1$... (1)

Differentiating with respect to x ,

$$\frac{2x}{a^2} + \frac{2y}{b^2 + \lambda} \frac{dy}{dx} = 0$$

$$\frac{2y}{b^2 + \lambda} \frac{dy}{dx} = -\frac{2x}{a^2}$$

$$\frac{dy}{dx} = -\frac{x(b^2 + \lambda)}{ya^2} \quad \dots (2)$$

$$(1) \Rightarrow \frac{y^2}{b^2 + \lambda} = 1 - \frac{x^2}{a^2}$$

$$= \frac{a^2 - x^2}{a^2}$$

$$\frac{y^2}{a^2 - x^2} = \frac{b^2 + \lambda}{a^2}$$

Equation (2) becomes $\frac{dy}{dx} = -\frac{x}{y} \left(\frac{y^2}{a^2 - x^2} \right)$

$$\frac{dy}{dx} = -\frac{xy}{a^2 - x^2}$$

Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$, we get

$$-\frac{dx}{dy} = -\frac{xy}{a^2 - x^2}$$

$$\frac{dy}{dx} = \frac{a^2 - x^2}{xy}$$

Integrating on both sides, we get $\int y dy = \int \frac{a^2 - x^2}{x} dx$

$$\frac{y^2}{2} = a^2 \log x - \frac{x^2}{2} + C$$

$$\frac{y^2}{2} + \frac{x^2}{2} = a^2 \log x + C$$

$x^2 + y^2 = 2a^2 \log x + 2C$ which is the required orthogonal trajectories.

Example 5: Find the orthogonal trajectories of the series of hypocycloids $x^{2/3} + y^{2/3} = a^{2/3}$

Solution :

The given equation is $x^{2/3} + y^{2/3} = a^{2/3}$

Differentiating with respect to x, we get

$$\frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3} \frac{dy}{dx} = 0$$

$$\frac{2}{3}y^{-1/3} \frac{dy}{dx} = -\frac{2}{3}x^{-1/3}$$

$$\frac{dy}{dx} = -\frac{x^{-1/3}}{y^{-1/3}} = -\frac{y^{1/3}}{x^{1/3}}$$

Replacing $\frac{dy}{dx}$ by $-\frac{dx}{dy}$, we get

$$-\frac{dx}{dy} = -\frac{y^{1/3}}{x^{1/3}}$$

$$\frac{dy}{dx} = \frac{x^{1/3}}{y^{1/3}}$$

Integrating on both sides, we get $\int y^{1/3} dy = \int x^{1/3} dx$

$$\frac{y^{4/3}}{4/3} = \frac{x^{4/3}}{4/3} + C$$

$y^{4/3} - x^{4/3} = C$ is the required orthogonal trajectories.

Example 6: Find the orthogonal trajectories of the curve of the family $2x^2 + y^2 = kx$

Solution:

The given equation is $2x^2 + y^2 = kx$... (1)

Differentiating with respect to x,

$$4x + 2y \frac{dy}{dx} = k \quad \dots (2)$$

To eliminate k, divide (2) by (1), we get

$$\frac{4x + 2y \frac{dy}{dx}}{2x^2 + y^2} = \frac{1}{x}$$

$$(i.e.) (4x + 2y \frac{dy}{dx})x = 2x^2 + y^2$$

$$4x^2 + 2xy \frac{dy}{dx} = 2x^2 + y^2$$

$$\frac{dy}{dx} = \frac{y^2 - 2x^2}{2xy}$$

Replacing $\frac{dy}{dx}$ by $-\frac{dx}{dy}$, we get

$$-\frac{dx}{dy} = \frac{y^2 - 2x^2}{2xy}$$

$$\frac{dy}{dx} = -\frac{2xy}{y^2 - 2x^2} \quad \dots(3)$$

This is a homogeneous equation.

put $y = vx$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

Equation (3) becomes $v + x \frac{dv}{dx} = -\frac{2vx^2}{v^2x^2 - 2x^2}$

$$x \frac{dv}{dx} = \frac{2v}{2 - v^2} - v = \frac{2v - 2v + v^3}{2 - v^2}$$

$$x \frac{dv}{dx} = \frac{v^3}{2 - v^2}$$

Integrating on both sides, we get

$$\int \frac{dx}{x} = \int \left(\frac{2}{v^3} - \frac{1}{v} \right) dv$$

$$\log x = -\frac{1}{v^2} - \log v + \log C$$

$$\log x + \log v - \log C = -\frac{1}{v^2}$$

$$\log\left(\frac{vx}{C}\right) = -\frac{1}{v^2}$$

$$\text{(i.e.) } \log\left(\frac{y}{C}\right) = -\frac{x^2}{y^2}$$

$$x^2 = -y^2 \log\left(\frac{y}{C}\right), \text{ which is the required orthogonal trajectories.}$$

Example 7: Find the orthogonal trajectories of the family of ellipses $\frac{x^2}{a^2} + \frac{y^3}{4a^2} = 1$

Solution:

$$\text{The given equation is } \frac{x^2}{a^2} + \frac{y^2}{4a^2} = 1$$

$$\frac{4x^2 + y^2}{4a^2} = 1$$

$$\text{(i.e.) } 4x^2 + y^2 = 4a^2 \quad \dots(1)$$

Differentiating with respect to x,

$$4(2x) + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{4x}{y}$$

Replacing $\frac{dy}{dx} = -\frac{dx}{dy}$, we get

$$-\frac{dx}{dy} = -\frac{4x}{y}$$

$$\frac{dy}{dx} = \frac{y}{4x}$$

Integrating on both sides, we get $4 \int \frac{dy}{y} = \int \frac{dx}{x}$

$$4 \log y = \log x + \log C$$

$y^4 = Cx$ is the required orthogonal trajectories.

Example 8: Find the orthogonal trajectories of the system of circles $r = a \cos \theta$

Solution:

The given equation is $r = a \cos \theta$... (1)

Differentiating with respect to θ , we get

$$\frac{dr}{d\theta} = -a \sin \theta \quad \dots (2)$$

To eliminate a, divide (2) by (1)

$$\frac{\frac{dr}{d\theta}}{r} = -\frac{\sin \theta}{\cos \theta}$$

$$\frac{dr}{d\theta} = -r \tan \theta$$

Replacing $\frac{dr}{d\theta}$ by $-r^2 \frac{d\theta}{dr}$

$$-r^2 \frac{d\theta}{dr} = -r \tan \theta$$

$$\tan \theta = r \frac{d\theta}{dr}$$

Integrating on both sides, we get

$$\int \frac{dr}{r} = \int \frac{d\theta}{\tan \theta}$$

$$\log r = \log \sin \theta + \log C$$

$r = C \sin \theta$ which is the required orthogonal trajectories.

Example 9: Find the O.T of the cardioids $r = a(1 - \cos \theta)$, where ‘a’ being a parameter.

Solution:

The given equation is $r = a(1 - \cos \theta)$, ...(1)

Differentiating with respect to θ , we get

$$\frac{dr}{d\theta} = a \sin \theta$$
...(2)

To eliminate a, divide (2) by (1)

$$\frac{\frac{dr}{d\theta}}{r} = \frac{\sin \theta}{1 - \cos \theta}$$

$$\frac{dr}{d\theta} = \frac{r \sin \theta}{1 - \cos \theta}$$

Replacing $\frac{dr}{d\theta}$ by $-r^2 \frac{d\theta}{dr}$

$$-r^2 \frac{d\theta}{dr} = \frac{r \sin \theta}{1 - \cos \theta}$$

Integrating on both sides, we get

$$\int \frac{dr}{r} = \int \frac{(1 - \cos \theta)}{\sin \theta} d\theta$$

$$\log r = -\int (\operatorname{cosec} \theta - \cot \theta) d\theta$$

$$\log r = \log(\operatorname{cosec} \theta + \cot \theta) + \log \sin \theta + \log c$$

$$\log r = \log \left(\frac{1}{\sin \theta} + \frac{\cos \theta}{\sin \theta} \right) \sin \theta \cdot c$$

$$r = \frac{1 + \cos \theta}{\sin \theta} (\sin \theta) c$$

$r = c(1 + \cos \theta)$ which is the required orthogonal trajectories.

Example 10: Find the orthogonal trajectories of the curves $r^n = a^n \cos n\theta$ and deduce the orthogonal trajectories of the series of lemniscates $r^2 = a^2 \cos 2\theta$

Solution:

The given equation is $r^n = a^n \cos n\theta$... (1)

Differentiating with respect to θ ,

$$nr^{n-1} \frac{dr}{d\theta} = -a^n n \sin n\theta$$

... (2)

To eliminate a^n , divide (2) by (1)

$$\frac{nr^{n-1} \frac{dr}{d\theta}}{r^n} = -n \frac{\sin n\theta}{\cos n\theta}$$

$$\frac{dr}{d\theta} = -r \tan n\theta$$

Replacing $\frac{dr}{d\theta}$ by $-r^2 \frac{d\theta}{dr}$

$$-r^2 \frac{d\theta}{dr} = -r \tan n\theta$$

$$\frac{dr}{d\theta} = \frac{r}{\tan n\theta}$$

Integrating on both sides, we get

$$\int \frac{dr}{r} = \int \cot n\theta \, d\theta$$

$$\log r = \frac{\log(\sin n\theta)}{n} + \log c$$

$$n \log r = \log(\sin n\theta) + n \log c$$

$$r^n = c^n \sin n\theta \quad \dots(3)$$

This is the required orthogonal trajectories.

To deduce the O.T of the series of lemniscates $r^2 = a^2 \cos 2\theta$ put $n = 2$ in (3), we get $r^2 = c^2 \sin 2\theta$ which is the required O.T

Example 11: Prove that the system of confocal and coaxial parabola $y^2 = 4a(x + a)$ is self orthogonal.

Solution :

The given equation is $y^2 = 4a(x + a)$ (1)

Differentiating with respect too x,

$$2y \frac{dy}{dx} = 4a$$

$$\frac{dy}{dx} = \frac{2a}{y}$$

$$a = \frac{y}{2} \frac{dy}{dx} \quad \dots(2)$$

Equation (1) becomes $y^2 = 4 \left(\frac{y}{2} \frac{dy}{dx} \right) \left(x + \frac{y}{2} \frac{dy}{dx} \right)$

$$= 2 \left(y \frac{dy}{dx} \right) \left(x + \frac{y}{2} \frac{dy}{dx} \right)$$

$$y^2 = 2xy \left(\frac{dy}{dx} \right) + y^2 \left(\frac{dy}{dx} \right)^2 \quad \dots(3)$$

Replacing $\frac{dy}{dx}$ by $-\frac{dx}{dy}$, we get

$$y^2 = 2xy \left(-\frac{dx}{dy} \right) + y^2 \left(-\frac{dx}{dy} \right)^2$$

$$y^2 = -2xy \left(\frac{dx}{dy} \right) + y^2 \left(\frac{dx}{dy} \right)^2$$

$$= -2xy \left(\frac{dx}{dy} \right) + y^2 \left(\frac{dx}{dy} \right)^2$$

Multiplying $\left(\frac{dy}{dx} \right)^2$, we get

$$y^2 \left(\frac{dy}{dx} \right)^2 = -2xy \left(\frac{dy}{dx} \right) + y^2$$

$\therefore y^2 = 2xy \left(\frac{dy}{dx} \right) + y^2 \left(\frac{dy}{dx} \right)^2$ which is the required orthogonal trajectories.

Since (3) is the same as (4), the system of confocal and coaxial parabolas is self orthogonal, (i.e.) each member of (1) cuts every other member orthogonally.

Exercises

1. Find the equation of the system of O.T of the family of curves $x^2 - y^2 = a^2$
2. Find the O.T. of the family of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{4a^2} = 1$
3. Find the equation to the O.T to the family of ellipses given by $\frac{x^2}{a^2} + \frac{y^2}{a^2 + \lambda} = 1$, where λ is an arbitrary constant.
4. Prove that the system of confocal and coaxial parabola $x^2 = 4a(y + a)$ is self orthogonal.
5. Find the equation of the system of O.T of a series of confocal and coaxial parabolas $r = \frac{2a}{1 + \cos \theta}$
6. Find the O.T of the family of semi cubical parabola $ay^2 = x^3$, where a is a variable parameter.
7. Show that the cardioids $r = a(1 + \cos \theta)$ and $r = b(1 - \cos \theta)$ intersect at right angles.
8. Find the orthogonal trajectories of the family of curves $r^2 = k(r \sin \theta - 1)$

Answers:

- | | |
|----------------------------------|--|
| 1. $xy = A$ | 5. $r = \frac{2c}{1 - \cos \theta}$ |
| 2. $y^4 = \frac{c}{x}$ | 6. $2x^2 + 3y^2 = C$ |
| 3. $x^2 + y^2 = 2a^2 \log x + C$ | 8. $r \sin \theta = 2 \tan \theta + C$ |

4.6. Applications of Differential Equations

The Law of Natural Growth

When a natural substance increases in magnitude as a result of some action which affects all parts equally, the rate of increase depends on the amount of the substance present. For instance, if N represents the total population of a country, its rate of increase with time is $\frac{dN}{dt}$. Under ideal conditions, granting all the necessities of a comfortable living to the people and assuming constant birth and death rates and in the absence of such factors as war, famine, disease, emigration, etc., one can reasonably guess that the rate of increase of the population will be proportional to the total population at any given time. Hence, the growth of population will obey a differential equation of the form $\frac{dN}{dt} = kN$... (1) where k is a positive constant. More generally, if N represents any self-generating population, the rate of growth of N will be represented by the same differential equation.

The solution of the equation (1) is $N = Ce^{kt}$ suppose $N = N_0$, when observations begin at $t = 0$, then $C = N_0$ and hence $N = N_0 e^{kt}$. As k is positive, N is steadily increasing.

Law of Decomposition (Decay)

In a radioactive substance, atoms disintegrate spontaneously. If at any time t , the number of radioactive atoms present is N , the rate at which N is decreasing is $\frac{dN}{dt}$ and this is negative. This rate of decrease or decay is found to be proportional to N itself. Hence we have the law, $\frac{dN}{dt} = -kN$, where k is a positive constant. As before, the solution of this equation is $N = N_0 e^{-kt}$, where N_0 is the number of radioactive atoms present at time $t = 0$. Due to the exponential factor with negative index, in this case N decreases steadily.

Cooling of Bodies

According to Newton's law of cooling, the rate at which the temperature T of a hot body decreases is proportional to the difference between the temperature of the body and the temperature T_0 of the surrounding air. Hence $\frac{dT}{dt} = -k(T - T_0)$

Put $T - T_0 = y$, then $\frac{dT}{dt} = \frac{dy}{dt}$. The above equation becomes $\frac{dy}{dt} = -ky$, which is of the same form as before. Here y is the excess of temperature of the body over that of the surrounding air at time t .

Simple Electric Circuits

We shall consider circuits made up of

- (i) three passive elements – resistance, inductance, capacitance and
- (ii) an active element – voltage source which may be a battery or generator

R (ohms) - Resistance

L – Inductance

C – Capacitance

E – Electromotive force

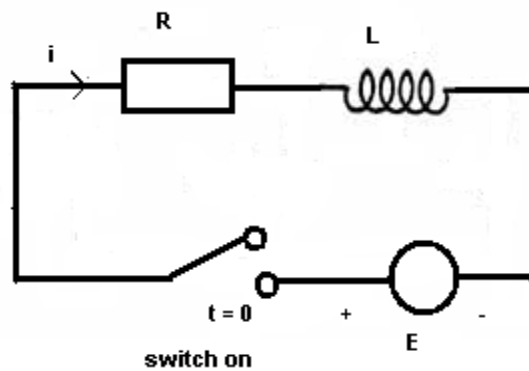


Fig. 4.3

Resistance: Resistance is that circuit parameter, which opposes the flow of current, $E = IR$ is the voltage drop across a resistance R.

Inductance : Inductance is that circuit parameter which opposes a change in current. $E = L \frac{dI}{dt}$, that is the voltage drop across an inductance L is the rate of change of current in times.

Capacitance : Capacitance is that circuit parameter which opposes a change in voltage $e = \frac{q}{c}$,

since the current is the rate of flow of positive charge $i = \frac{dq}{dt}$.

Kirchhoff's Law

(i) In a closed circuit the sum of the voltage drops across each element of the circuit is equal to the impressed voltage.

(ii) At a junction of two or more elements of a circuit, the amount of current into the junction point is equal to the amount of current away from the junction point. $L \frac{dI}{dt} + RI = E$ (or)

$L \frac{di}{dt} + Ri = E$, where I is the current if E is constant or a function of t.

Unimolecular Chemical Reactions

In certain simple chemical reactions, the rate at which the molecules of a substance 'A' undergo a chemical change is proportional to the number of molecules of it, which are present at any particular instant. Hence, if N is the number of molecules of A, remaining at time t, then

$\frac{dN}{dt} = -kN$ which is of the same form as before. Such a reaction is known as a first-order process.

Example 1: The number N of bacteria in a culture grew at a rate proportional to N . The value of N was initially 100 and increased to 332 in one hour. What was the value of N after $1\frac{1}{2}$ hours?

Solution :

The differential equation to be solved is $\frac{dN}{dt} = kN$... (1)

The solution of (1) is $N = Ce^{kt}$... (2)

When $t = 0$, $N = 100$, so $C = 100$, $N = 100e^{kt}$... (3)

When $t = 3600$ secs, $N = 332$ (i.e.) $332 = 100e^{3600k}$... (4)

When $t = \frac{3}{2}$ hours = 5400 secs, $N = 100e^{5400k}$... (5)

From (4), $e^{3600k} = \frac{332}{100}$

Hence $(e^{3600k})^{\frac{3}{2}} = \left(\frac{332}{100}\right)^{\frac{3}{2}}$ (i.e.) $e^{5400k} = \left(\frac{332}{100}\right)^{\frac{3}{2}}$

Putting in 5, $N = 100\left(\frac{332}{100}\right)^{\frac{3}{2}} = 605$

Example 2: In a certain chemical reaction, the rate of conversion of a substance at time t is proportional to the quantity of the substance still untransformed at that instant. At the end of one hour 60 grams remain and at the end of four hours 21 grams. How many grams of the first substance were there initially?

Solution :

Let x grams of the substance remain after t hours. Then the differential Equation of the reaction is

$$\frac{dx}{dt} = -kx \dots (1) \text{ where } k \text{ is a positive constant.}$$

$$\text{The solution of (1) is } x = Ce^{-kt} \dots (2)$$

$$\text{When } t = 1, \quad x = 60 \text{ so } 60 = Ce^{-k} \dots (3)$$

$$\text{When } t = 4, \quad x = 21 \text{ so } 21 = Ce^{-4k} \dots (4)$$

Initially when $t = 0$, $x = C$

To find C , eliminate k from (3) and (4).

$$\text{From (3) } e^{-k} = \frac{60}{C}$$

Substituting in equation (4), we get

$$21 = C \left(\frac{60}{C} \right)^4 = \frac{60^4}{C^3}$$

$$\text{(i.e.) } C^3 = \frac{60^4}{21}$$

Taking logarithms,

$$3 \log C = 4 \log 60 - \log 21 = 5.7906$$

$$\log C = 1.9302$$

$$C = \text{anti log } 1.9302 = 85.15$$

Hence, initially there were 85 grams of the substance.

Example 3: The current i in an electric circuit containing an inductance L and a resistance R in series are acted on by an electromotive force $E \sin wt$. Find the value of the current at any time t , if initially no current in the circuit. (Use Kirchhoff's law). (Jan-2011)

Solution :

By Kirchhoff's law, $L \frac{di}{dt} + Ri = E \sin wt$

$$\frac{di}{dt} + \frac{R}{L}i = \frac{E \sin wt}{L}$$

This is linear differential equation.

$$\therefore \text{I.F.} = e^{\int \frac{R}{L} dt} = e^{\frac{Rt}{L}}$$

$\therefore$ The solution is given by $i (\text{I.F.}) = \int \frac{E}{L} \sin wt \cdot e^{\frac{Rt}{L}} + C$

$$i e^{\frac{Rt}{L}} = \frac{E}{L} \int e^{\frac{Rt}{L}} \sin wt dt + C$$

$$i e^{\frac{Rt}{L}} = \frac{E}{L} \left[\frac{e^{\frac{Rt}{L}}}{\frac{R^2}{L^2} + w^2} \left(\frac{R}{L} \sin wt - w \cos wt \right) \right] + C$$

$$\left[\therefore \int e^{ax} \sin bxdx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) \right]$$

$$i = \frac{E}{L} \left[\frac{L^2}{R^2 + L^2 w^2} \left(\frac{R}{L} \sin wt - w \cos wt \right) \right] + c e^{-\frac{Rt}{L}} \quad \dots(1)$$

Initially when $t = 0$, $i = 0$

$$\therefore 0 = \frac{-ELw}{R^2 + L^2 w^2} + C$$

$$C = \frac{ELw}{R^2 + L^2w^2}$$

$$\begin{aligned} \therefore (1) \text{ becomes } i &= \frac{EL}{R^2 + L^2w^2} \left[\left(\frac{R}{L} \sin wt - w \cos wt \right) \right] + \frac{ELw}{R^2 + L^2w^2} e^{-\frac{Rt}{L}} \\ &= E \left[\left(\frac{R}{R^2 + L^2w^2} \sin wt - \frac{Lw}{R^2 + L^2w^2} w \cos wt \right) + \frac{Lw}{R^2 + L^2w^2} e^{-\frac{Rt}{L}} \right] \end{aligned}$$

$$\text{Let } \sin \phi = \frac{Lw}{\sqrt{R^2 + L^2w^2}}, \quad \cos \phi = \frac{R}{\sqrt{R^2 + L^2w^2}}$$

$$\therefore i = \frac{E}{\sqrt{R^2 + L^2w^2}} \left[(\cos \phi \sin wt - \sin \phi \cos wt) + \sin \phi e^{-\frac{Rt}{L}} \right]$$

$$i = \frac{E}{\sqrt{R^2 + L^2w^2}} \left[\sin(wt - \phi) + \sin \phi e^{-\frac{Rt}{L}} \right] \text{ which gives the current at any time } t.$$

Example 4: When a switch is closed in a circuit containing a battery E , a resistance R and an inductance L the current i builds up at a rate given by $L \frac{di}{dt} + Ri = E$, find i as a function of t .

Solution:

$$\text{Given that } L \frac{di}{dt} + Ri = E$$

$$\text{(i.e.) } \frac{di}{dt} + \frac{R}{L}i = \frac{E}{L}$$

This is a linear equation. $\therefore$ I.F. = $e^{\int \frac{R}{L} dt} = e^{\frac{Rt}{L}}$

$\therefore$ The solution is $i(I.F.) = \int \frac{E}{L}(I.F.)dt + c$

$$ie^{\frac{Rt}{L}} = \int \frac{E}{L} e^{\frac{Rt}{L}} dt + C$$

$$= \frac{E}{L} \frac{e^{\frac{Rt}{L}}}{\frac{R}{L}} + C$$

$$ie^{\frac{Rt}{L}} = \frac{E}{R} e^{\frac{Rt}{L}} + C$$

$$i = \frac{E}{R} + Ce^{-\frac{Rt}{L}}$$

Example 5: According to Newton's law of cooling, the rate at which a substance cools in moving air is proportional to the difference between the temperature of the substance and that of the air. If the temperature of the air is 30°C and the substance cools from 100°C to 70°C in 15 minutes, find when the temperature will be 40°C .

Solution :

Let the unit of time be a minute and T the temperature of the substance at any instant t . Then by

Newton's law of cooling, we have $\frac{dT}{dt} = -k(T - 30)$ (or) $\frac{dT}{T - 30} = -kdt$

Integrating on both sides, we get

$$\log(T - 30) = -kt + C \quad \dots(1)$$

Initially, when $t = 0$, $T = 100$, $C = \log 70$

$\therefore$ (1) becomes, $\log(T - 30) = -kt + \log 70$

$$kt = \log 70 - \log(T - 30) \quad \dots(2)$$

Also, when $t = 15$, $T = 70$

$$15k = \log 70 - \log 40 \quad \dots(3)$$

Dividing (2) by (3),

$$\frac{t}{15} = \frac{\log 70 - \log 10}{\log 70 - \log 40} = \frac{\log_e (7)}{\log_e \left(\frac{7}{4}\right)} = 3.48$$

$$t = 15(3.48) = 52.2 \text{ minutes.}$$

Hence the temperature will be 40°C after 52.2 minutes.

Exercise

1. Radium decomposes at a rate proportional to the amount present. If half the original quantity disappears in 1600 years, find what percentage of the original amount disappears in 200 years?

2. In a certain chemical reaction, the rate of conversion of a substance A at time t is proportional to the quantity of A remaining untransformed at the instant. If that amount is 45 at the end of one hour and 15 at end of 3 hours, find the amount at the start.

3. The equation of the electromotive force in terms of the current i for an electric circuit having a resistance R and a condenser of capacity C is given by $\frac{di}{dt} + \frac{i}{Rc} = \frac{1}{R} \cdot \frac{dE}{dt}$

Find the current i when E = a constant.

Answers

$$1. 8.3\% \quad 2. 40\sqrt{\frac{3}{2}} \quad 3. i = \frac{E}{R} e^{-\frac{t}{Rc}}$$

4.6.1. Geometrical Applications

Let P be a point (x, y) on a curve. PT and PN are the tangent and the normal at P , meeting the x axis at T and N respectively. PM is the ordinate of P . Let $\angle PTM = \psi$. We know that

$\tan \psi = \frac{dy}{dx}$. The several lines have been given the following names; PT is the tangent; TM the subtangent; PN the normal and MN the subnormal. The following formulae express these lengths in terms of $\frac{dy}{dx}$

$$1. \text{ Sub tangent} = TM = \frac{y}{\left(\frac{dy}{dx}\right)} = y \frac{dx}{dy};$$

$$2. \text{ Sub normal} = MN = y \frac{dy}{dx}$$

$$3. \text{ tangent} = PT = y \sqrt{1 + \left(\frac{dx}{dy}\right)^2};$$

$$4. \text{ Normal} = PN = y \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

Also, if ds is a small element of the arc of the curve at P, we have the relation

$$\left(\frac{ds}{dx}\right)^2 = 1 + \left(\frac{dy}{dx}\right)^2$$

We can setup easily the differential equation of a family of curves, given some geometrical property which can be expressed in terms of the tangent and normal lines.

Example 1: Find the equation of the curves for which the subnormal at each point has a constant value a.

Solution :

Length of the subnormal at any point $(x,y) = y \frac{dy}{dx}$

Hence $y \frac{dy}{dx} = a$ is the differential equation to the given family

(i.e.) $ydy = adx$

Integrating, $y^2 = 2ax + C$ which represents a family of parabolas.

Example 2: The part of the normal between the point $p(x,y)$ of a curve and the x-axis is a constant k . Form the differential equation of the family and show that the curves are all circles.

Solution :

The equation of normal is given by $y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = k$... (1)

This is the differential equation of the family.

Squaring (1) on both sides, we get $y^2 \left[1 + \left(\frac{dy}{dx}\right)^2\right] = k^2$

$$(i.e.) \left(\frac{dy}{dx}\right)^2 = \frac{k^2}{y^2} - 1 = \frac{k^2 - y^2}{y^2}$$

$$\frac{dy}{dx} = \pm \frac{\sqrt{(k^2 - y^2)}}{y}$$

Separating the variables, $\frac{ydy}{\sqrt{(k^2 - y^2)}} = \pm dx$

Integrating on both sides, we get $\sqrt{(k^2 - y^2)} = \pm x + c$

$$k^2 - y^2 = (c \pm x)^2$$

(i.e.) $(x \pm c)^2 + y^2 = k^2$ which is clearly the equation to a family of circles.

Exercise

1. Find the equation of the curve in which the subnormal is constant and equal to 5 and which passes through the point (2, 1)
2. Find the curves for which the subnormal at any point on it is the square root of the abscissa.
3. Determine the curve in which the subtangent is equal to the square of the subnormal and which passes through the point (1,1)

Answers

1. $y^2 = 10x - 19$
2. $y^2 = \frac{4}{3}x^{3/2} + C$
3. $3y^{4/3} = 4x - 1$

4.7. Equations not of first degree

Introduction: In this chapter, we shall consider the differential equations of the first order but with a degree higher than the first. For convenience, we denote $\frac{dy}{dx}$ by p . The general differential equation of the first order and degree n is of the form

$$p^n + P_1 p^{n-1} + P_2 p^{n-2} + \dots + P_{n-1} p + P_n = 0 \dots (1) \text{ where } P_1, P_2, \dots, P_n \text{ are functions of } x \text{ and } y.$$

To solve this equation (1), we consider the following cases :

1. Equations solvable for p
2. Equations solvable for y
3. Equations solvable for x

4.7.1. Equations solvable for p

Suppose in (1), the left side can be resolved into n rational factors of the first degree. Then (1) can be put in the form $(p - F_1)(p - F_2)\dots(p - F_n) = 0 \dots (2)$ where $F_1, F_2, \dots, F_n$ are functions of x and y . Equating each factor of (2) to zero, we get n differential equations each of the first

order and first degree. They are $\frac{dy}{dx} = F_1, \frac{dy}{dx} = F_2, \dots \frac{dy}{dx} = F_n$. These equations can be solved by the usual methods. Let solutions of these equations be respectively. $f_1(x, y, c_1) = 0, f_2(x, y, c_2) = 0, \dots f_n(x, y, c_n) = 0 \dots (3)$ where $c_1, c_2, \dots, c_n$ are arbitrary constants. The n solutions in (3) can be combined into one single solution of the form $f_1(x, y, c) \times f_2(x, y, c) \times \dots \times f_n(x, y, c) = 0 \dots (4)$

Since every particular solution obtained from equations (3) can also be obtained from equation (4) by giving suitable value of c , there is no loss of generality in replacing the n arbitrary constants $c_1, c_2, \dots, c_n$ by a single arbitrary constant c .

Example 1: Solve $p^2 - 9p + 18 = 0$, **where** $p = \frac{dy}{dx}$

Solution :

The given equation is $p^2 - 9p + 18 = 0$

$$(p-6)(p-3) = 0$$

$$p-6=0 \quad p-3=0$$

$$p=6 \quad p=3$$

$$\frac{dy}{dx} = 6 \quad \frac{dy}{dx} = 3$$

$$\int dy = 6 \int dx \quad \int dy = 3 \int dx$$

$$y = 6x + c, \quad y = 3x + c$$

The combined solution is $(y-6x-c)(y-3x-c) = 0$

Example 2: Solve $p^2 + 2py \cot x = y^2$

(Nov- 2003,2007)

Solution :

The given equation is $p^2 + 2py \cot x - y^2 = 0$ solving for y,

$$p = \frac{-2y \cot x \pm \sqrt{4y^2 \cot^2 x + 4y^2}}{2} = \frac{-2y \cot x \pm \sqrt{4y^2(1 + \cot^2 x)}}{2}$$

$$= \frac{-2y \cot x \pm 2y \sqrt{\cos^2 x}}{2} = \frac{-2y \cot x \pm 2y \cos x}{2}$$

$$p = -y \cot x \pm y \cos x$$

Case (i) $p = -y \cot x + y \cos x$

$$\frac{dy}{dx} = y(\cos x - \cot x)$$

Integrating, $\log y = -\log(\cos x + \cot x) - \log \sin x + \log c$

$$(i.e.) y(\cos x + \cot x) \sin x = c$$

$$(i.e.) y \left(\frac{1}{\sin x} + \frac{\cos x}{\sin x} \right) \sin x = c$$

$$(i.e.) y(1 + \cos x) = c$$

Case (ii) $p = -y \cot x - y \cos x$

$$\frac{dy}{dx} = y(-\cot x - \cos x)$$

Integrating, $\int \frac{dy}{y} = \int (-\cot x - \cos x) dx$

$$\log y = -\log \sin x + \log(\cos x + \cot x) + \log c$$

$$(i.e.) \ y = c \left(\frac{\cos ecx + \cot x}{\sin x} \right) = c \left(\frac{1 + \cos x}{\sin^2 x} \right)$$

$$= c \frac{(1 + \cos x)}{1 - \cos^2 x} = \frac{c}{1 - \cos x}$$

$$(i.e.) \ y(1 - \cos x) = c$$

The combined solution is $y(1 \pm \cos x) = c$

Example 3: Solve $p^3 + 2xp^2 - y^2 p^2 - 2xy^2 p = 0$ **where** $p = \frac{dy}{dx}$

Solution :

The given equation is $p(p^2 + 2xp - y^2 p - 2xy^2) = 0$

$$(i.e.) \ p(p(p + 2x) - y^2(p + 2x)) = 0$$

$$(i.e.) \ p(p + 2x)(p - y^2) = 0$$

$$p = 0, \ p + 2x = 0, \ p - y^2 = 0$$

$$\frac{dy}{dx} = 0, \ \frac{dy}{dx} = -2x, \ \frac{dy}{dx} = y^2$$

Integrating the above equations, we get

$$y = c, \int dy = -2 \int x dx, \ \int \frac{dy}{y^2} = \int dx$$

$$y = -x^2 + c \quad -\frac{1}{y} = x + c$$

$$y + x^2 - c = 0, \ yx + cy + 1 = 0$$

The combined solution is $(y - c)(y + x^2 - c)(xy + cy + 1) = 0$

Example 4: Solve $x^2 \frac{d^2 y}{dx^2} - 2xy \frac{dy}{dx} + 2y^2 = x^2$

Solution :

The given equation is $x^2 p^2 - 2xyp + 2y^2 - x^2 = 0$

$$\begin{aligned} p &= \frac{2xy \pm \sqrt{4x^2 y^2 - 4(x^2)(2y^2 - x^2)}}{2x^2} \\ &= \frac{2xy \pm \sqrt{-4x^2 y^2 + 4x^4}}{2x^2} = \frac{2xy \pm 2x\sqrt{x^2 - y^2}}{2x^2} \\ p &= \frac{y \pm \sqrt{x^2 - y^2}}{x} \end{aligned} \quad \dots(1)$$

Let $y = vx$

$$\frac{dy}{dx} = p = v + \frac{xdv}{dx}$$

$$\begin{aligned} \text{Equation (1) becomes, } v + x \frac{dv}{dx} &= \frac{vx \pm \sqrt{x^2 - v^2 x^2}}{x} \\ &= v \pm \sqrt{1 - v^2} \\ \pm \frac{dv}{\sqrt{1 - v^2}} &= \frac{dx}{x} \end{aligned}$$

$$\text{Case (i) } \int \frac{dv}{\sqrt{1 - v^2}} = \int \frac{dx}{x}$$

$$\sin^{-1} v = \log x + c$$

$$\sin^{-1} \left(\frac{y}{x} \right) = \log x + c$$

Case (ii) $\int \frac{-dv}{\sqrt{1-v^2}} = \int \frac{dx}{x}$

$$-\sin^{-1} v = \log x + c_2$$

$$-\sin^{-1} \left(\frac{y}{x} \right) = \log x + c_2$$

$$\sin^{-1} \left(\frac{y}{x} \right) + \log x + c_2 = 0$$

Hence the combined solution is $\left(\sin^{-1} \left(\frac{y}{x} \right) - \log x - c \right) \left(\log x + \sin^{-1} \left(\frac{y}{x} \right) + c \right) = 0$

Example 5: Solve the equation $xp^2 - 2py + x = 0$

(May-2012, Jan-2013)

Solution :

The given equation is $xp^2 - 2py + x = 0$

Solving for p, $p = \frac{2y \pm \sqrt{4y^2 - 4x^2}}{2x}$

$$\frac{dy}{dx} = \frac{y}{x} \pm \sqrt{\left(\frac{y}{x} \right)^2 - 1} \quad \dots(1)$$

This is a homogeneous equation

Put $y = vx$

$$\frac{dy}{dx} = p = v + x \frac{dv}{dx}$$

Equation (1) becomes $v + x \frac{dv}{dx} = v \pm \sqrt{v^2 - 1}$

$$\text{(i.e.) } \frac{dv}{\sqrt{v^2 - 1}} = \pm \frac{dx}{x}$$

Integrating, we get

$$\int \frac{dv}{\sqrt{v^2-1}} = \pm \int \frac{dx}{x}$$

$$\log(v + \sqrt{v^2-1}) = \pm \log x + \log c$$

$$\therefore \frac{y + \sqrt{y^2 - x^2}}{x^2} = c \quad \text{and} \quad y + \sqrt{y^2 - x^2} = c$$

$$\text{(i.e.) } y + \sqrt{y^2 - x^2} - cx^2 = 0 \quad \text{and} \quad y + \sqrt{y^2 - x^2} - c = 0$$

$$\therefore \text{The combined solution is } (y + \sqrt{y^2 - x^2} - cx^2) (y + \sqrt{y^2 - x^2} - c) = 0$$

Example 6: Solve the equation $p(p+y) = x(x+y)$

Solution :

The given equation is $p^2 + yp - (x^2 + xy) = 0$

$$p = \frac{-y \pm \sqrt{y^2 + 4(x^2 + xy)}}{2} = \frac{-y \pm \sqrt{(y+2x)^2}}{2}$$

$$\therefore p = \frac{-y \pm (y+2x)}{2}$$

Case (i) $p = \frac{-y + y + 2x}{2} = x$

$$\frac{dy}{dx} = x$$

Integrating on both sides, we get $\int dy = \int x dx$

$$y = \frac{x^2}{2} + \frac{c}{2}$$

$$\text{(i.e.) } 2y - x^2 - c = 0$$

$$\text{Case(ii) } p = \frac{-y - y - 2x}{2}$$

$$\frac{dy}{dx} = -x - y$$

$$\frac{dy}{dx} + y = -x, \text{ which is a linear equation.}$$

$$\therefore \text{ I.F.} = e^x$$

$$\therefore \text{ The solution is } y(\text{I.F.}) = \int -x(\text{I.F.})dx + c$$

$$ye^x = \int -xe^x dx + c$$

$$= -xe^x + e^x + c$$

$$y + x - 1 - ce^{-x} = 0$$

Hence, the combined solution is $(2y - x^2 - c)(y + x - 1 - ce^{-x}) = 0$

$$\text{Example 7: Solve } p - \frac{1}{p} = \frac{x}{y} - \frac{y}{x}$$

(Nov- 2004)

Solution:

$$\text{The given equation is } \frac{p^2 - 1}{p} = \frac{x^2 - y^2}{xy}$$

$$p^2 xy - xy = px^2 - py^2$$

$$p^2 xy - xy - px^2 + py^2 = 0$$

$$p^2 xy + p(y^2 - x^2) - xy = 0$$

$$\begin{aligned}
 p &= \frac{(x^2 - y^2) \pm \sqrt{(y^2 - x^2)^2 + 4x^2y^2}}{2xy} \\
 &= \frac{(x^2 - y^2) \pm \sqrt{y^4 + x^4 + 2x^2y^2}}{2xy} \\
 &= \frac{(x^2 - y^2) \pm \sqrt{(x^2 + y^2)^2}}{2xy} = \frac{(x^2 - y^2) \pm (x^2 + y^2)}{2xy}
 \end{aligned}$$

Case(i) $p = \frac{(x^2 - y^2) + (x^2 + y^2)}{2xy}$

$$\frac{dy}{dx} = \frac{x^2 - y^2 + x^2 + y^2}{2xy} = \frac{2x^2}{2xy}$$

$$\frac{dy}{dx} = \frac{x}{y}$$

Integrating on both sides, we get $\int y dy = \int x dx$

$$\frac{y^2}{2} = \frac{x^2}{2} + c$$

$$y^2 - x^2 = 2c$$

Case (ii) $p = \frac{(x^2 - y^2) - (x^2 + y^2)}{2xy}$

$$\frac{dy}{dx} = \frac{-2y^2}{2xy} = -\frac{y}{x}$$

Integrating on both sides, we get $\int \frac{dy}{y} = -\int \frac{dx}{x}$

$$\log y = -\log x + \log c$$

$$xy - c = 0$$

Hence the combined solution is $(y^2 - x^2 - c)(xy - c) = 0$

Example 2: Solve : $2p^2 - (x + 2y^2)p + xy^2 = 0$

Solution:

The given equation is $2p^2 - (x + 2y^2)p + xy^2 = 0$

$$\begin{aligned} p &= \frac{(x + 2y^2) \pm \sqrt{(x + 2y^2)^2 - 8xy^2}}{4} \\ &= \frac{(x + 2y^2) \pm \sqrt{x^2 + 4y^4 + 4xy^2 - 8xy^2}}{4} \\ &= \frac{(x + 2y^2) \pm \sqrt{x^2 + 4y^4 - 4xy^2}}{4} \\ &= \frac{(x + 2y^2) \pm \sqrt{(x - 2y^2)^2}}{4} = \frac{(x + 2y^2) \pm (x - 2y^2)}{4} \end{aligned}$$

Case (i) $p = \frac{x + 2y^2 + x - 2y^2}{4}$

(i.e.) $\frac{dy}{dx} = \frac{x}{2}$

Integrating on both sides, we get $\int 2dy = \int xdx$

$$2y = \frac{x^2}{2} + c$$

$$4y - x^2 - 2c = 0$$

Case (ii) $p = \frac{x + 2y^2 - x + 2y^2}{4}$

(i.e.) $\frac{dy}{dx} = y^2$

Integrating on both sides, we get $\int \frac{dy}{y^2} = \int dx$

$$-\frac{1}{y} = x + c_2$$

$$\frac{1}{y} + x + c_2 = 0$$

Hence the combined solution is $(4y - x^2 - c)\left(\frac{1}{y} + x + c\right) = 0$

Exercise

Solve the following equations

1. $p^2 - 5p + 6 = 0$

2. $x^2 p^2 + 3xyp + 2y^2 = 0$

3. $xyp^2 + p(3x^2 - 2y^2) - 6xy = 0$

4. $xy\left(\frac{dy}{dx}\right)^2 - (x^2 + y^2)\frac{dy}{dx} + xy = 0$

5. $4p^2 - 8p + 3 = 0$

6. $xy^2(p^2 + 2) - 2py^2 - x^2 = 0$

Answers

1. $(y - 3x - c)(y - 2x - c) = 0$

2. $(xy - c)(x^2 - y - c) = 0$

3. $(y^2 + 2x^2 - c)(y - cx^2) = 0$

4. $(y - cx)(y^2 - x^2 - c) = 0$

5. $(2y - x - c)(2y - 3x - c) = 0$

6. $(x^2 - y^2 - c)(x^2 - y^2 + cx^4) = 0$

4.7.2. Equations solvable for y

If $F(x, y, p) = 0$ is the differential equation of the first order but of degree higher than one, then y can be expressed explicitly as a single valued function of x and p.

(i.e.) The equation of this type can be re-written as $y = f(x, p)$... (1)

Differentiating (1) with respect to x, we get $p = \phi\left(x, p, \frac{dp}{dx}\right)$... (2)

Equation (2) is a differential equation of the first order and first degree in the variables x and p. It can be solved by the usual methods. Let the solution of (2) be $\psi(x, p, c) = 0$... (3) where c is an arbitrary constant. If we eliminate p between (1) and (3), the eliminant is the general solution of the given equation. If p cannot be easily eliminated between (1) and (3), they jointly provide the required solution in terms of the parameter p.

Example 1: Solve $y = -px + x^4 p^2$

Solution:

The given equation is $y = -px + x^4 p^2$... (1)

Differentiating (1) with respect to 'x'

$$\frac{dy}{dx} = -\left[p + x \frac{dp}{dx}\right] + \left[x^4 \cdot 2p \cdot \frac{dp}{dx} + p^2 \cdot 4x^3\right]$$

$$p = -p - x \frac{dp}{dx} + 2px^4 \frac{dp}{dx} + 4p^2 x^3$$

$$2p + x \frac{dp}{dx} - 2px^4 \frac{dp}{dx} - 4p^2 x^3 = 0$$

$$x \frac{dp}{dx} (1 - 2px^3) + 2p(1 - 2px^3) = 0$$

$$\left[x \frac{dp}{dx} + 2p\right] [1 - 2px^3] = 0$$

Case (i) Let $x \frac{dp}{dx} + 2p = 0$

$$\frac{xdp}{dx} = -2p$$

Integrating on both sides, we get $\int \frac{dp}{p} = -2 \int \frac{dx}{x}$

$$\log p = -2 \log x + \log c$$

$$\log p + 2 \log x = \log c$$

$$px^2 = c$$

$$p = \frac{c}{x^2} \quad \dots(2)$$

Substituting (2) in (1), we get $y = -\frac{c}{x^2} \cdot x + x^4 \frac{c^2}{x^4}$

$$y = -\frac{c}{x} + c^2 \text{ which is the general solution.}$$

Case (i) Let $1 - 2px^3 = 0$

$$2px^3 = 1$$

$$p = \frac{1}{2x^3} \quad \dots(3)$$

Substituting (3) in (1), we get

$$y = -\frac{1}{2x^3} x + x^4 \frac{1}{4x^6}$$

$$y = -\frac{1}{2x^2} + \frac{1}{4x^2}$$

$y = -\frac{1}{4x^2}$ which is also a solution of the equation (1). However, it does not contain any arbitrary constant. This solution is called a singular solution.

Example 2: Solve : $4y = x^2 + p^2$

(Apr- 2003, 2007)

Solution :

The given equation is $4y = x^2 + p^2$... (1)

Differentiating (1) with respect to x, we get

$$4 \frac{dy}{dx} = 2x + 2p \frac{dp}{dx}$$

$$(i.e.) \quad 4p = 2x + 2p \frac{dp}{dx}$$

$$\frac{dp}{dx} = \frac{2p - x}{p}$$

To solve this, put $p = vx$

$$\frac{dp}{dx} = v + x \frac{dv}{dx}$$

$$\text{Hence } v + x \frac{dv}{dx} = \frac{2vx - x}{vx} = \frac{2v - 1}{v}$$

$$x \frac{dv}{dx} = \frac{2v - 1}{v} - \frac{v}{1} = \frac{2v - v^2 - 1}{v}$$

$$\frac{v dv}{2v - v^2 - 1} = \frac{dx}{x}$$

$$\frac{v}{(v-1)^2} dv = -\frac{dx}{x}$$

$$\frac{(v-1+1)}{(v-1)^2} dv = -\frac{dx}{x}$$

$$\frac{1}{v-1} dv + \frac{1}{(v-1)^2} dv = -\frac{dx}{x}$$

Integrating on both sides, we get $\log(v-1) - \frac{1}{v-1} = -\log x + c$

$$\text{(i.e.) } \log(v-1)x = \frac{1}{v-1} + c$$

$$\text{Put } v = \frac{p}{x}, \text{ we get } \log\left(\frac{p}{x} - 1\right)x = \frac{1}{\frac{p}{x} - 1} + c$$

$$\log(p-x) = \frac{x}{p-x} + c \quad \dots(2)$$

$$\text{The given equation is } 4y = x^2 + p^2 \quad \dots(3)$$

Equation (2) and (3) which express x and y in terms of p constitute the solution.

Example 3: Solve : $y = 3x + \log p$

Solution :

$$\text{The given equation is } y = 3x + \log p \quad \dots(1)$$

Differentiating (1) with respect to x, we get $\frac{dy}{dx} = 3 + \frac{1}{p} \frac{dp}{dx}$

$$p = 3 + \frac{1}{p} \frac{dp}{dx}$$

$$p - 3 = \frac{1}{p} \frac{dp}{dx}$$

$$1 = \frac{dp}{dx} \cdot \frac{1}{p(p-3)} \quad \dots(2)$$

By resolving $\frac{1}{p(p-3)}$ into partial fraction, we have $\frac{1}{p(p-3)} = \frac{A}{p} + \frac{B}{p-3}$

$$1 = A(p-3) + B(p)$$

$$\text{Put } p = 0 ; 1 = A(-3) \Rightarrow A = -\frac{1}{3}$$

$$\text{Put } p = 3 ; 1 = B(3) \Rightarrow B = \frac{1}{3}$$

$$\therefore (2) \text{ becomes } 1 = \frac{dp}{dx} \left(-\frac{1}{3p} + \frac{1}{3} \cdot \frac{1}{p-3} \right)$$

$$-\frac{1}{3} \int \frac{dp}{p} + \frac{1}{3} \int \frac{1}{p-3} dp = \int dx$$

$$-\frac{1}{3} \log p + \frac{1}{3} \log(p-3) = x + c$$

$$\text{(i.e.) } \log \frac{p-3}{p} = 3x + c$$

$$\text{(i.e.) } \frac{p-3}{p} = e^{3x+c},$$

$$\text{(i.e.) } 1 - \frac{3}{p} = e^{3x} c_1 \quad (\because c_1 = e^c)$$

$$\text{(i.e.) } 1 - c_1 e^{3x} = \frac{3}{p}$$

$$\text{(i.e.) } p = \frac{3}{1 - c_1 e^{3x}} \quad \dots(3)$$

Substituting (3) in (1), we get $y = 3x + \log \frac{3}{1 - c_2 e^{3x}}$

This gives the general solution of the given equation.

Example 4: Solve $y = x + a \tan^{-1} p$

Solution:

The given equation is $y = x + a \tan^{-1} p$... (1)

Differentiating (1) with respect to x,

$$\frac{dy}{dx} = 1 + a \cdot \frac{1}{p^2 + 1} \frac{dp}{dx}$$

$$p = 1 + \frac{a}{p^2 + 1} \frac{dp}{dx}$$

$$p - 1 = \frac{a}{p^2 + 1} \frac{dp}{dx}$$

$$\frac{dx}{a} = \frac{dp}{(p^2 + 1)(p - 1)} \quad \dots (2)$$

$$\frac{1}{(p^2 + 1)(p - 1)} = \frac{A}{p - 1} + \frac{Bp + C}{p^2 + 1} \quad (\text{using partial fraction})$$

$$1 = A(p^2 + 1) + (Bp + C)(p - 1)$$

$$\text{Put } p = 1, \quad 1 = A(2) + 0 \Rightarrow A = \frac{1}{2}$$

$$\text{Put } p = 0, \quad 1 = A + C(-1)$$

$$1 = A - C$$

$$1 = \frac{1}{2} - C \Rightarrow C = -\frac{1}{2}$$

$$\text{Put } p = -1, 1 = A(2) + (-B + C)(-2)$$

$$1 = 2A + 2B - 2C$$

$$1 = 1 + 2B + 1$$

$$2B = -1 \Rightarrow B = -\frac{1}{2}$$

$$\therefore (2) \text{ becomes, } dp \left[\frac{1}{2(p-1)} - \frac{p}{2(p^2+1)} - \frac{1}{2(p^2+1)} \right] = \frac{dx}{a}$$

$$\frac{1}{2} \int \frac{1}{p-1} dp - \frac{1}{2} \int \frac{p}{p^2+1} dp - \frac{1}{2} \int \frac{1}{p^2+1} dp = \int \frac{dx}{a}$$

$$\frac{1}{2} \left[\log(p-1) - \frac{1}{2} \log(p^2+1) - \tan^{-1} p \right] = \frac{x}{a} + c$$

$$x = \frac{a}{2} \left[\log \frac{p-1}{\sqrt{p^2+1}} - \tan^{-1} p \right] + c \quad \dots(3)$$

Substituting (3) in (1), we get

$$y = \frac{a}{2} \left[\log(p-1) - \frac{1}{2} \log(p^2+1) - \tan^{-1} p \right] + a \tan^{-1} p + c \quad \dots(4)$$

Since the elimination of p is very difficult, equation (3) and (4) gives the general solution.

Example 5: Solve $y = (1+p)x + p^2$

Solution :

The given equation is $y = (1+p)x + p^2$...(1)

Differentiating with respect to x ,

$$\frac{dy}{dx} = 1 + p + x \frac{dp}{dx} + 2p \frac{dp}{dx}$$

$$p = 1 + p + (x + 2p) \frac{dp}{dx}$$

$$(x + 2p) \frac{dp}{dx} + 1 = 0$$

$$(x + 2p) \frac{dp}{dx} = -1$$

$$x + 2p = -\frac{dx}{dp}$$

$$\frac{dx}{dp} + x = -2p \quad \dots(2)$$

This is a linear equation in x, $I.F = e^{\int dp} = e^p$

$\therefore$ The solution of (2) is $xe^p = \int -2pe^p dp + c$

$$= -2(pe^p - e^p) + c$$

$$x = 2 - 2p + ce^{-p} \quad \dots(3)$$

Substituting (3) in (1)

$$\begin{aligned} y &= (1 + p)(2 - 2p + ce^{-p}) + p^2 \\ &= 2 - 2p + ce^{-p} + 2p - 2p^2 + cpe^{-p} + p^2 \end{aligned}$$

Example 6: Solve $p^2x - 2py + ax = 0$

(Apr- 2010)

Solution :

The given equation is $p^2x - 2py + ax = 0$

$\dots(1)$

The given equation can be written as

$$y = \frac{p^2 x + ax}{2p} = \frac{1}{2} \left(px + \frac{ax}{p} \right)$$

$$\frac{dy}{dx} = p = \frac{1}{2} \left[p + x \frac{dp}{dx} + \frac{\left(pa - ax \frac{dp}{dx} \right)}{p^2} \right]$$

$$2p = p + x \frac{dp}{dx} + \frac{a \left(p - x \frac{dp}{dx} \right)}{p^2}$$

$$p = \frac{xp^2 \frac{dp}{dx} + a \left(p - x \frac{dp}{dx} \right)}{p^2}$$

$$p^3 = xp^2 \frac{dp}{dx} + ap - x \frac{dp}{dx} a$$

$$x \frac{dp}{dx} (p^2 - a) - p(p^2 - a) = 0$$

$$\left(x \frac{dp}{dx} - p \right) (p^2 - a) = 0$$

$$\therefore x \frac{dp}{dx} - p = 0 \quad \left[\because (p^2 - a) \neq 0 \right]$$

$$x \frac{dp}{dx} = p$$

$$\text{Integrating, } \int \frac{dp}{p} = \int \frac{dx}{x}$$

$$\log p = \log x + \log c$$

$$p = cx \quad \dots(2)$$

Substituting (2) in given equation,

$$c^2x^3 - 2cxy + ax = 0$$

$$2cxy = c^2x^3 + ax$$

$$2y = cx^2 + \frac{a}{c}$$

$$y = 2 - p^2 + c(1 + p)e^{-p} \quad \dots(3)$$

Since the elimination of p is very difficult, equation (2) and (3) gives the general solution.

Example 7: Solve $y = 2px - p^2$

Solution :

The given equation is $y = 2px - p^2 \quad \dots(1)$

Differentiating with respect to x ,

$$\frac{dy}{dx} = 2p + 2x \frac{dp}{dx} - 2p \frac{dp}{dx}$$

$$(i.e.) \quad p = 2p + (2x - 2p) \frac{dp}{dx}$$

$$p + (2x - 2p) \frac{dp}{dx} = 0$$

$$(i.e.) \quad 2x - 2p = -p \frac{dx}{dp}$$

$$p \frac{dx}{dp} + 2x - 2p = 0$$

$$\frac{dx}{dp} + \frac{2}{p}x = 2 \quad \dots(2)$$

which is linear equation

$$I.F = e^{\int \frac{2}{p} dp} = e^{2 \log p} = p^2$$

∴ The solution of (2) is

$$x(I.F) = \int 2(I.F) dp + c$$

$$xp^2 = \int 2p^2 dp + c$$

$$xp^2 = \frac{2}{3} p^3 + c$$

$$x = \frac{2}{3} p + cp^{-2} \quad \dots(3)$$

Substituting (3) in (1), we get $y = 2p \left(\frac{2}{3} p + cp^{-2} \right) - p^2$

$$y = \frac{4}{3} p^2 + 2cp^{-1} - p^2 \quad \dots(4)$$

Since the elimination of p is very difficult, equation (3) and (4) together constitute the general solution of (1)

Exercise

Solve the following differential equations:

1. $x + xp^2 - 2yp = 0$

4. $y - 2px = \tan^{-1}(xp^2)$

2. $x - yp = ap^2$

5. $p^2x - 2py - x = 0$

3. $y = 2p + 3p^2$

6. $y = x + p^2 - 2p$

Answers

1. $y = \frac{1}{2c} + \frac{x^2c}{2}$

2. $x = \frac{p}{\sqrt{1-p^2}} (c + a \sin^{-1} p)$

3. $x = 2 \log p + 6p + c$ with the given relation

$$5. c^2 x^2 = 2cy + 1$$

$$6. (x + c)^2 = 4(y + c)$$

$$4. y = 2\sqrt{cx} + \tan^{-1} c$$

4.7.3. Equations solvable for x

If $F(x, y, p) = 0$ be the differential equation of the first order but of degree higher than one, then x can be expressed explicitly as a single valued function of y and p.

(i.e.) The equation of this type can be re-written as $x = f(y, p)$... (1)

Differentiating (1) with respect to y, we get $\frac{1}{p} = \phi\left(y, p, \frac{dp}{dy}\right)$... (2)

Equation (2) is a differential Equation of the first order and first degree in the variables y and p. It can be solved by methods already known.

Let the solution of (2) be $\psi(y, p, c) = 0$... (3)

where c is an arbitrary constant

If we eliminate p between (1) and (3), the eliminant is the general solution of the given equation. If p cannot be easily eliminated between (1) and (3), they jointly provide the required solution in terms of the parameter p.

Note: Some differential equations can be put in both the forms $y = f(x, p)$ and $x = f(y, p)$. Both these forms may lead to the required solution. Sometimes one of the forms will lead to the required solution more easily than the other.

Example 1: Solve $y = 2px + y^2 p^3$

(Apr- 2010, 2011)

Solution :

Given equation is $y = 2px + y^2 p^3$... (1)

Solving for x, we have $x = \frac{1}{2} \left(\frac{y}{p} - y^2 p^2 \right)$

Differentiating bothsides with respect to y

$$\frac{dx}{dy} = \frac{1}{p} = \frac{1}{2} \left[\frac{1}{p} - \frac{y}{p^2} \frac{dp}{dy} - 2yp^2 - 2y^2 p \frac{dp}{dy} \right]$$

$$\text{(i.e.) } \frac{2}{p} = \frac{1}{p} - \frac{y}{p^2} \frac{dp}{dy} - 2yp^2 - 2y^2 p \frac{dp}{dy}$$

Multiplying by p^2 , we get

$$2p = p - y \frac{dp}{dy} - 2yp^4 - 2y^2 p^3 \frac{dp}{dy}$$

$$p + 2yp^4 + y \frac{dp}{dy} + 2y^2 p^3 \frac{dp}{dy} = 0$$

$$p(1 + 2yp^3) + y \frac{dp}{dy} (1 + 2yp^3) = 0$$

$$\left(p + y \frac{dp}{dy} \right) (1 + 2yp^3) = 0$$

Discarding the factor $(1 + 2yp^3)$,

$$p + y \frac{dp}{dy} = 0 \quad , \quad p = -y \frac{dp}{dy}$$

$$\frac{dy}{y} + \frac{dp}{p} = 0$$

Integrating, $\log y + \log p = \log c$

$$py = c \quad \text{ie. } p = \frac{c}{y} \quad \dots(2)$$

Substituting (2) in (1), we get $y = \frac{2cx}{y} + \frac{c^3}{y}$ ie. $y^2 = 2cx + c^3$, which is the required solution.

Example 2: Solve $p = \tan\left(x - \frac{p}{1+p^2}\right)$

Solution :

Given equation can be written as $x = \tan^{-1} p + \frac{p}{1+p^2} \quad \dots(1)$

Differentiating with respect to 'y'

$$\frac{dx}{dy} = \frac{1}{p} = \frac{1}{1+p^2} \frac{dp}{dy} + \frac{(1+p^2) - 2p^2}{(1+p^2)^2} \frac{dp}{dy}$$

$$\frac{1}{p} = \left[\frac{(1+p^2) + (1+p^2) - 2p^2}{(1+p^2)^2} \right] \frac{dp}{dy}$$

$$\frac{1}{p} = \frac{2}{(1+p^2)^2} \frac{dp}{dy}$$

$$\int dy = \int \frac{2p}{(1+p^2)^2} dp$$

$$y = c - \frac{1}{1+p^2} \quad \dots(2)$$

Since the elimination of p is very difficult, equations (1) and (2) together constitute the general solution.

Example 3: Solve $x = y + a \log p$

Solution :

The given equation is $x = y + a \log p$ (1)

Differentiating (1) with respect to y,

$$\frac{dx}{dy} = 1 + a \frac{1}{p} \frac{dp}{dy}$$

$$\frac{1}{p} = 1 + a \frac{1}{p} \frac{dp}{dy}$$

$$\frac{1}{p} - 1 = \frac{a}{p} \frac{dp}{dy}$$

$$1 - p = a \frac{dp}{dy}$$

$$dy = \frac{a}{1-p} dp$$

Integrating, $\int dy = -a \int \frac{dp}{p-1}$

$$y = -a \log(p-1) + c$$
(2)

Substituting (2) in (1), we get

$$x = -a \log(p-1) + a \log p + c$$

$$x = a[\log p - \log(p-1)] + c$$

$$x = a \log \frac{p}{p-1} + c$$
(3)

Since the elimination of p is very difficult, equations (2) and (3) give the general solution.

Example 4: Solve $p^3 - 4xyp + 8y^2 = 0$

(Apr- 2009)

Solution :

The given equation is $p^3 - 4xyp + 8y^2 = 0$ (1)

$$\text{Solving for } x, \quad 4x = \frac{p^2}{y} + \frac{8y}{p}$$

Differentiating with respect to y,

$$4 \frac{dx}{dy} = -\frac{p^2}{y^2} + \frac{2p}{y} \frac{dp}{dy} + \frac{8}{p} - \frac{8y}{p^2} \frac{dp}{dy}$$

$$\frac{4}{p} = -\frac{p^2}{y^2} + \left(\frac{2p}{y} - \frac{8y}{p^2} \right) \frac{dp}{dy} + \frac{8}{p}$$

$$\left(\frac{2p}{y} - \frac{8y}{p^2} \right) \frac{dp}{dy} = \frac{p^2}{y^2} - \frac{4}{p}$$

$$\frac{(2p^3 - 8y^2)}{yp^2} \frac{dp}{dy} = \frac{(p^3 - 4y^2)}{py^2}$$

$$2(p^3 - 4y^2)y \frac{dp}{dy} = p(p^3 - 4y^2)$$

$$(p^3 - 4y^2) \left[2y \frac{dp}{dy} - p \right] = 0$$

$$p^3 - 4y^2 = 0 \quad \text{or} \quad 2y \frac{dp}{dy} - p = 0$$

Case (i): Let $p^3 - 4y^2 = 0$

$$p^3 = 4y^2 \quad \dots(2)$$

Substituting (2) in (1),

$$4y^2 - 4xyp + 8y^2 = 0$$

$$4xyp = 12y^2$$

$$p = \frac{3y}{x}$$

$$p^3 = \frac{27y^3}{x^3} \quad \dots(3)$$

Equating (2) and (3), we get $4y^2 = \frac{27y^3}{x^3}$

(i.e.) $4x^3 = 27y$, which is the singular solution of (1)

Case (ii): Let $2y \frac{dp}{dy} - p = 0$

$$2y \frac{dp}{dy} = p \quad (\text{i.e.}) \quad 2 \frac{dp}{p} = \frac{dy}{y}$$

Integrating, we get $2 \log p = \log y + \log c$

$$(\text{i.e.}) \quad p^2 = cy \quad \dots(4)$$

Substituting (4) in (1),

$$pcy - 4xyp + 8y^2 = 0$$

$$(\text{i.e.}) \quad py(c - 4x) = -8y^2$$

Cancelling y and squaring, we get

$$p^2(c - 4x)^2 = 64y^2$$

$$cy(c - 4x)^2 = 64y^2$$

$$c(c - 4x)^2 = 64y, \text{ which is the general solution.}$$

Example 5: Solve $x = y + p^2$

Solution :

The given equation is $x = y + p^2$

Differentiating with respect to 'y'

$$\frac{dx}{dy} = \frac{1}{p} = 1 + 2p \frac{dp}{dy}$$

$$\frac{1}{p} - 1 = 2p \frac{dp}{dy}$$

$$\frac{1-p}{p} = 2p \frac{dp}{dy}$$

$$\frac{dy}{2} = \frac{p^2}{1-p} dp$$

$$\int \frac{p^2}{1-p} dp = \frac{1}{2} \int dy$$

$$\int \frac{p^2 - 1 + 1}{1-p} dp = \frac{1}{2} \int dy$$

$$-\int \frac{p^2 - 1 + 1}{p-1} = \frac{1}{2} \int dy$$

$$-\left[\int \frac{(p-1)(p+1)}{(p-1)} dp + \int \frac{dp}{p-1} \right] = \frac{1}{2} \int dy$$

$$-\frac{p^2}{2} - p - \log(p-1) = \frac{y}{2} + c$$

$$2 \left[-\frac{p^2}{2} - p - \log(p-1) \right] + c = y$$

$$y = -p^2 - 2p - 2\log(p-1) + c$$

Since the elimination of p is very difficult,

$$x = -[2p + 2\log(p-1) + c]$$

$y = -p^2 - 2p - 2\log(p-1) + c$ are the general solution.

Example 6: Solve $x = p + p^4$

Solution :

The given equation is $x = p + p^4$... (1)

Differentiating (1) with respect to y,

$$\frac{dx}{dy} = \frac{dp}{dy} + 4p^3 \frac{dp}{dy}$$

$$\frac{1}{p} = \frac{dp}{dy} (1 + 4p^3)$$

$$dy = dp(1 + 4p^3)p$$

$$\int dy = \int (p + 4p^4) dp$$

$$y = \frac{p^2}{2} + \frac{4p^5}{5} + c \quad \dots (2)$$

Since the elimination of p is very difficult, equations (1) and (2) give the general solution.

Example 7: Solve the equation $p^3 - 2xyp + 4y^2 = 0$

Solution :

The given equation is $p^3 - 2xyp + 4y^2 = 0$... (1)

$$\text{(i.e.) } 2x = \frac{p^2}{y} + \frac{4y}{p} \quad \dots (2)$$

Differentiating (2) with respect to y

$$2 \frac{dx}{dy} = \frac{2}{p} = -\frac{p^2}{y^2} + \frac{2p}{y} \frac{dp}{dy} + \frac{4}{p} - \frac{4y}{p^2} \frac{dp}{dy}$$

$$\text{(i.e.) } \left(\frac{2p}{y} - \frac{4y}{p^2} \right) \frac{dp}{dy} + \left(\frac{2}{p} - \frac{p^2}{y^2} \right) = 0$$

$$\text{(i.e.) } \frac{2}{p^2 y} (p^3 - 2y^2) \frac{dp}{dy} - \frac{1}{py^2} (p^3 - 2y^2) = 0$$

$$\text{(i.e.) } (p^3 - 2y^2) \left(2y \frac{dp}{dy} - p \right) = 0$$

$$p^3 - 2y^2 = 0 \quad \text{or} \quad 2y \frac{dp}{dy} - p = 0$$

Case(i): $2y \frac{dp}{dy} = p$

$$\int \frac{2dp}{p} = \int \frac{dy}{y}$$

$$2 \log p = \log y + \log c$$

$$p^2 = cy \quad \dots(3)$$

Substituting (3) in (1), we get

$$cpy - 2xyp + 4y^2 = 0$$

$$p(c - 2x) = -4y$$

Squaring and using (4), $cy(c - 2x)^2 = 16y^2$

$$\text{(i.e.) } c(c - 2x)^2 = 16y \quad \text{which is the general solution of (1)}$$

Case (ii): $p^3 - 2y^2 = 0$

$$p^3 = 2y^2 \quad \dots(4)$$

Substituting (4) in (1)

$$2y^2 - 2xyp + 4y^2 = 0$$

$$(i.e.) \quad xp = 3y$$

Raising power three on both sides,

$$x^3 p^3 = 27y^3$$

$$(i.e.) \quad x^3(2y^2) = 27y^3 \quad (\because \text{from 4})$$

$2x^3 = 27y$, which is the singular solution of (1).

Exercise

Solve the following equations

$$1. \quad y = 3px + 6p^2y^2$$

$$3. \quad y = 2px + p^2y$$

$$2. \quad y^2 \log y = xyp + p^2$$

$$4. \quad p^2y + 2px - y = 0$$

Answers

$$1. \quad y^3 = 3cx + 6c^2$$

$$3. \quad y^2 = 2cy + c^2$$

$$2. \quad \log y = cx + c^2$$

$$4. \quad cx = \log y - c^2$$

4.7.4. Clairaut's Equation

An equation of the form $y = px + f(p)$ is called Clairaut's equation.

Let the Clairaut's equation be $y = px + f(p)$... (1)

Differentiating (1), with respect to x,

$$p = p + [x + f'(p)] \frac{dp}{dx}$$

$$[x + f'(p)] \frac{dp}{dx}$$

$$\frac{dp}{dx} = 0 \quad \dots(2)$$

$$\text{or } f'(p) + x = 0 \quad \dots(3)$$

$$\text{Solving (2), we get } p = c \quad \dots(4)$$

By Eliminating p between (1) and (4), we get the general solution of (1) as $y = cx + f(c)$

Thus the general solution of a Clairaut's equation is obtained by replacing p by c in the given equation. Eliminating p between (1) and (3), we also get a solution of (1).

This solution does not contain any arbitrary constant. Also it cannot be obtained as a particular case of the general solution. This solution is called the singular solution of (1).

Note: The singular solution of (1) is the eliminant of p between $y = px + f(p)$ and

$x + \frac{df}{dp} = 0$. Equivalently, the singular solution of (1) is the eliminant of c between

$y = cx + f(c)$ and $x + \frac{df}{dc} = 0$. Equivalently, the singular solution of (1) is the eliminant of p

between $y = cx + f(c)$ and $x + \frac{df}{dc} = 0$.

Example 1: Solve $y = (x-a)p - p^2$

(Nov-2006, May- 008)

Solution :

The given equation is $y = px - ap - p^2 \quad \dots(1)$

This is Clairaut's equation.

Differentiating (1) with respect to x,

$$\frac{dy}{dx} = p \cdot 1 + x \frac{dp}{dx} - a \frac{dp}{dx} - 2p \frac{dp}{dx}$$

$$\text{(i.e.) } p = p + x \frac{dp}{dx} - a \frac{dp}{dx} - 2p \frac{dp}{dx}$$

$$\text{or } \frac{dp}{dx}(x - a - 2p) = 0$$

$$\text{Either } \frac{dp}{dx} = 0 \quad \text{or} \quad x - a - 2p = 0$$

$$\text{Case (i): Let } \frac{dp}{dx} = 0$$

$$\text{Integrating, we get } p = c \quad \dots(2)$$

Substituting (2) in (1), we get

$$y = cx - ac - c^2$$

This is the general solution

$$\text{Case(ii): Let } x - a - 2p = 0$$

$$\text{Then } p = \frac{x - a}{2}$$

$$\text{Put } p = \frac{x - a}{2} \text{ in (1)}$$

$$\begin{aligned} y &= (x - a) \left(\frac{x - a}{2} \right) - \frac{(x - a)^2}{4} \\ &= \frac{(x - a)^2}{2} - \frac{(x - a)^2}{4} = \frac{(x - a)^2}{4} \end{aligned}$$

$$\therefore (x - a)^2 = 4y, \text{ which is the singular solution.}$$

Example 2: Solve : $y = px + ap^{-1}$

Solution :

The given equation is $y = px + ap^{-1}$... (1)

This is Clairaut's equation.

Put $p = c$ in (1), we get

$$y = cx + ac^{-1} \quad \dots (2)$$

This is general solution

Differentiating (2) with respect to c ,

$$0 = x + a \left(-\frac{1}{c^2} \right)$$

$$0 = x - \frac{a}{c^2}$$

$$x = \frac{a}{c^2} \quad \therefore c = \sqrt{\frac{a}{x}} \quad \dots (3)$$

Substituting (3) in (1), we get

$$y = x \sqrt{\frac{a}{x}} + a \sqrt{\frac{a}{x}}$$

$$y = \sqrt{ax} + \sqrt{ax} = 2\sqrt{ax}$$

$$\therefore y^2 = 4ax$$

This is the singular solution.

Example 3: Solve $y = px + \sqrt{1 + p^2}$ **(Nov - 2005)**

Solution :

The given equation is $y = px + \sqrt{1 + p^2}$... (1)

This is Clairaut's equation

$$\text{Put } p = c \text{ in (1), } y = cx + \sqrt{1+c^2} \quad \dots(2)$$

This is general solution

Differentiating (2) with respect to 'c',

$$0 = x + \frac{1}{2\sqrt{1+c^2}} 2c = x + \frac{c}{\sqrt{1+c^2}}$$

$$x = -\frac{c}{\sqrt{1+c^2}} \Rightarrow x^2 = \frac{c^2}{1+c^2} \quad \dots(3)$$

$$(2) \Rightarrow y = c \left(\frac{-c}{\sqrt{1+c^2}} \right) + \sqrt{1+c^2} = \frac{-c^2 + 1 + c^2}{\sqrt{1+c^2}} = \frac{1}{\sqrt{1+c^2}}$$

$$y^2 = \frac{1}{1+c^2} \quad \dots(4)$$

Adding (3) and (4), we get,

$$x^2 + y^2 = \frac{c^2}{1+c^2} + \frac{1}{1+c^2}$$

$$x^2 + y^2 = 1$$

This is the singular solution.

Example 4: Solve $y = (x-1)p + \tan^{-1} p$

Solution :

$$\text{The given equation is } y = (x-1)p + \tan^{-1} p \quad \dots(1)$$

This is Clairaut's equation

$$\text{Put } p = c \text{ in (1),}$$

$$y = (x-1)c + \tan^{-1}(c) \quad \dots(2)$$

This is general solution.

Differentiating (2) with respect to 'c',

$$0 = (x-1) + \frac{1}{1+c^2}$$

$$\frac{1}{1+c^2} = -x+1$$

$$1+c^2 = \frac{1}{1-x}$$

$$c^2 = \frac{1}{1-x} - 1 = \frac{1+x-1}{1-x} = \frac{x}{1-x}$$

$$c = \sqrt{\frac{x}{1-x}}$$

Substituting c value in (2)

$$y = (x-1)\sqrt{\frac{x}{1-x}} + \tan^{-1} \sqrt{\frac{x}{1-x}}$$

This is singular solution.

Example 5: Solve $y = 2px - p^2$

Solution :

The given equation is $y = 2px - p^2$... (1)

This is Clairaut's equation

Put $p = c$ in (1), we get $y = 2cx - c^2$... (2)

This is general solution.

Differentiating (2) with respect to c,

$$0 = 2x - 2c$$

$$2c = 2x \Rightarrow c = x$$

Substituting the value of 'c' in (2), we get

$$y = 2x^2 - x^2 = x^2$$

This is the singular solution

Example 6: Solve $x^2(y - px) = yp^2$ **(Apr- 2003, Nov-2011)**

Solution :

The given equation is $x^2(y - px) = yp^2$... (1)

Put $x^2 = X$, and $y^2 = Y$

Then $2xdx = dX$ and $2ydy = dY$

$$\frac{dY}{dX} = P = \frac{y}{x} \frac{dy}{dx} = \frac{y}{x} p$$

$$\text{or } p = \frac{x}{y} P$$

Equation (1) becomes

$$X \left(y - \frac{x^2}{y} P \right) = y \frac{x^2}{y^2} P^2$$

$$X(y^2 - x^2 P) = x^2 P^2$$

$$X(Y - XP) = XP^2$$

$$Y = XP + P^2$$

This is Clairaut's equation.

So the general solution is $Y = Xc + c^2$

$$\text{(i.e.) } y^2 = x^2 c + c^2 \quad \dots (2)$$

Differentiating (2) with respect to c

$$0 = x^2 + 2c$$

$$c = -\frac{x^2}{2}$$

Substituting the value of 'c' in (2), we get

$$y^2 = x^2 \left(-\frac{x^2}{2} \right) + \frac{x^4}{4} = -\frac{x^4}{2} + \frac{x^4}{4}$$

$$y^2 = -\frac{x^4}{4}$$

$$4y^2 = -x^4$$

This is the singular solution

Example 7: Solve $(px - y)(py + x) = 2p$

(Jan - 2010)

Solution :

Given equation is $(px - y)(py + x) = 2p$... (1)

Put $X = x^2$, $Y = y^2$

$$dX = 2x dx \quad dY = 2y dy$$

$$\therefore \frac{dY}{dX} = \frac{y}{x} \frac{dy}{dx} \quad (\text{i.e.}) \quad P = \frac{y}{x} p \text{ (say)}$$

$$\text{or } p = \frac{x}{y} P$$

Substituting p value in (1),

$$\left(\frac{x}{y} Px - y \right) \left(\frac{x}{y} Py + x \right) = 2 \frac{x}{y} P$$

$$\left(\frac{x^2}{y}P - y\right)(xP + x) = 2\frac{x}{y}P$$

$$x\left(\frac{x^2P - y^2}{y}\right)(P + 1) = 2\frac{x}{y}P$$

$$x^2P - y^2 = \frac{2P}{P + 1}$$

$$y^2 = x^2P - \frac{2P}{P + 1}$$

$$Y = XP - \frac{2P}{P + 1} \quad \dots(2)$$

This is Clairaut's equation.

Put $P = c$ in (2), we get

$$Y = Xc - \frac{2c}{c + 1}$$

$$\text{(i.e.) } y^2 = x^2c - \frac{2c}{c + 1}$$

This is the general solution.

Example 8: Solve $y = 2px + yp^2$ **(May - 2012)**

Solution:

The given equation is $y = 2px + yp^2$...(1)

$$\text{Put } y^2 = Y$$

$$2y \frac{dy}{dx} = \frac{dY}{dx}$$

$$2yp = P(\text{say}) \Rightarrow y^2 p^2 = \frac{P^2}{4}$$

Multiplying throughout the given equation by y, it becomes,

$$y^2 = 2ypx + y^2 p^2 \quad \dots(2)$$

$$(\text{i.e.}) Y = Px + \frac{P^2}{4} \quad \dots(3)$$

which is a Clairaut's equation.

$$\therefore \text{The general solution of (3) is } Y = cx + \frac{c^2}{4} \quad \dots(4)$$

Differentiating (4) partially with respect to c,

$$0 = x + \frac{c}{2} \quad \dots(5)$$

Eliminating between (4) and (5), we get

$$Y = -x^2$$

(i.e.) $x^2 + y^2 = 0$, which is the singular solution of (1)

Exercise

Solve the following:

$$1. \ y = px + \frac{a}{p}$$

$$2. \ y = px + ap(1-p)$$

$$3. \ p = \log(px - y)$$

$$4. \ y = 2px + y^2 p^2$$

$$5. \ (px - y)(py + x) = h^2 p$$

$$6. \ y = xp^2 + p$$

Answers

1. $y = cx + \frac{a}{c}$ is general solution, $y^2 = 4ax$ is the singular solution.
2. $y = cx + ac - ac^2$ is general solution, $(x + a)^2 = 4ay$ is the singular solution.
3. $y = cx - e^c$ is general solution, $y = x \log x - x$ is the singular solution.
4. $y^2 = 2cx + c^2$ (Hint: Put $2x = X$, $y^2 = Y$)
5. $y^2 = cx^2 + \frac{h^2 c}{1 + c}$
6. $x(1 - p)^2 = c + \log p - p$ with the given relation.

PART A - QUESTIONS AND ANSWERS

1. Define exact differential equations.

Solution: Any differential equation of the form $Mdx + Ndy = 0$ is said to be exact, if it satisfies the condition that $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, where M, N being functions of x and y only.

2. What is the General solution of the exact differential equation?

Solution: The General solution of the exact differential equation is $\int Mdx + \int Ndy = C$,

Where $\int Ndy = \int$ (The terms not containing x)dy.

3. When do you say that the D.E is exact? (Jan - 2010)

Solution: Any D.E of the form $Mdx + Ndy = 0$ is said to be exact if it satisfies the

condition that $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

4. Check whether the following differential equation is exact or non-exact.

$$(a^2 - 2xy - y^2)dx - (x + y)^2 dy = 0$$

Solution: The given differential equation is $(a^2 - 2xy - y^2)dx - (x + y)^2 dy = 0 \dots (1)$

Equation (1) is of the form $Mdx + Ndy = 0$,

$$\text{Where } M = (a^2 - 2xy - y^2) \dots (2)$$

$$N = -(x + y)^2 \dots (3)$$

$$\frac{\partial M}{\partial y} = -2x - 2y \dots (4)$$

$$\frac{\partial N}{\partial x} = -2x - 2y \dots (5)$$

From equations (4) and (5) we get $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Therefore, the given differential equation is exact.

5. Define Integrating factor.

Solution: If an equation of the form $Mdx + Ndy = 0$ is not exact, it can always be made exact by multiplication with a proper factor. Such a multiplier is called an Integrating factor.

6. Solve $\left(\frac{dy}{dx}\right)^2 - 6\frac{dy}{dx} + 8 = 0$ **(Jan - 2011)**

Solution: Given equation can be written as $p^2 - 6p + 8 = 0$,

$$(p-2)(p-4)=0$$

$$(p-2)=0 \text{ \& } (p-4)=0$$

$$p=2$$

$$p=4$$

$$\frac{dy}{dx}=2$$

$$\frac{dy}{dx}=4$$

$$dy=2dx$$

$$dy=4dx$$

$$2dx-dy=0$$

$$4dx-dy=0$$

$$2 \int dx - \int dy = c_1 \quad 4 \int dx - \int dy = c_2$$

$$2x - y = c_1 \quad 4x - y = c_2$$

$$2x - y = c_1 \quad 4x - y - c_2 = 0$$

$$(2x - y - c_1) (4x - y - c_2) = 0$$

$\therefore$ The General solution is given by $(2x - y - c) (4x - y - c) = 0$.

7. Solve $p^2 - 7p + 12 = 0$, where $p = \frac{dy}{dx}$.

Solution: Given that $p^2 - 7p + 12 = 0$

$$(p-3)(p-4)=0$$

$$(p-3)=0 \quad (p-4)=0$$

$$(p-3)=0 \quad (p-4)=0$$

$$p = 3$$

$$p = 4$$

$$\frac{dy}{dx} = 3$$

$$\frac{dy}{dx} = 4$$

$$dy = 3dx$$

$$dy = 4dx$$

$$3dx - dy = 0$$

$$4dx - dy = 0$$

$$3 \int dx - \int dy = 0$$

$$4 \int dx - \int dy = 0$$

$$3x - y = c_1$$

$$4x - y = c_2$$

$$3x - y - c_1 = 0$$

$$4x - y - c_2 = 0$$

$\therefore$ The General solution is $(3x - y - c_1)(4x - y - c_2) = 0$

8. What is meant by General and Singular solution?

Solution: The solution of the differential equation of the first order contains the arbitrary constant is called **General solution**. The solution of the differential equation of the first order does not contain any arbitrary constant is called **Singular solution**.

9. Define Clairaut's form (or) Clairaut's equation and its solution.

(Jan-2011, Nov-2011)

Solution: The differential equations of the form $y = xp + f(p)$ (or)

$y = x \frac{dy}{dx} + f\left(\frac{dy}{dx}\right)$ is called Clairaut's Equation. Put $p=c$. Then the solution of

Clairaut's equation is $y = xc + f(c)$

10. What is the General solution of $y = px + ap$?

Solution: The Given equation is $y = px + ap$. This equation is of the form

$y = xp + f(p)$, which is known as Clairaut's form.

Put $p = c$, then we get the General solution.

$\therefore y = cx + ac$ is the General solution of the given equation (1).

11. Solve $y = (x-a)p - p^2$

(May-2012)

Solution: The given equation can be written as $y = xp - ap - p^2$

This equation is of the form $y = xp + f(p)$, which is known as Clairaut's form.

Put $p = c$, then the general solution is $y = xc - ac - c^2$

12. Define first order linear differential equation.

Solution: The first order differential equation is of the form $\frac{dy}{dx} + Py = Q$

Where P and Q are functions of x only (or) constant. It is called Linear differential equation.

13. What is the General solution of the first order linear differential equation?

Solution: The General solution of the first order linear differential equation $\frac{dy}{dx} + Py = Q$

$$\text{is } ye^{\int P dx} = \int Qe^{\int P dx} dx + c$$

14. Define Orthogonal Trajectory.

Solution: If two families of curves are such that each member of either family cuts each member of the other family at right angles, the members of one family are known as the orthogonal trajectory of the other.

15. What is the Orthogonal Trajectory of $x^2 + y^2 = a^2$?

Solution: Given that $x^2 + y^2 = a^2$... (1)

Differentiate equation (1) with respect to x ,

$$2x + 2y \frac{dy}{dx} = 0$$

$$\text{Put } \frac{dy}{dx} = -\frac{dx}{dy}$$

$$-\frac{x}{y} = -\frac{dx}{dy}$$

$$\frac{dy}{y} = \frac{dx}{x}$$

$$\frac{dx}{x} - \frac{dy}{y} = 0$$

$$\int \frac{dx}{x} - \int \frac{dy}{y} = \log c$$

$$\log x - \log y = \log c$$

$$\log\left(\frac{x}{y}\right) = \log c$$

$$\frac{x}{y} = c$$

$\therefore$ The Orthogonal trajectory of $x^2 + y^2 = a^2$ is $\frac{x}{y} = c$

16. Write down the standard form of Bernoulli's equation.

(Nov-2011)

Solution: The Bernoulli's equation is $\frac{dy}{dx} + Py = Qy^n$

The general solution is $y^{1-n} e^{\int p(1-n)dx} = \int (1-n)Qe^{\int p(1-n)dx} dx + c$

17. Solve: $xdx = y(x^2 + y^2 - 1)dy$

(Jan -2010)

Solution: The given differential equation is $xdx + ydy = y(x^2 + y^2)dy$

$$\frac{2xdx + 2ydy}{x^2 + y^2} = 2ydy$$

$$\frac{d(x^2 + y^2)}{x^2 + y^2} = 2ydy$$

Integrating on both sides we get,

$$\int \frac{d(x^2 + y^2)}{x^2 + y^2} = 2 \int \frac{y dy}{x^2 + y^2} + c$$

$$\log(x^2 + y^2) = y^2 + c.$$

18. Solve: $(2xy^2 - y)dx + xdy = 0$

(Jan - 2010)

Solution: The given differential equation is $(2xy^2 - y)dx + xdy = 0$

$$2x dx + \frac{xdy - ydx}{y^2} = 0$$

$$2x dx - \left(\frac{ydx - xdy}{y^2} \right) = 0$$

$$2x dx - d\left(\frac{x}{y}\right) = 0$$

Integrating on both sides we get,

$$\int 2x dx - \int d\left(\frac{x}{y}\right) = c$$

$$x^2 - \frac{x}{y} = c$$

$$x^2 y - x = cy.$$

19. Solve $x = y + a \log p$

(May- 2012)

Solution: Given $x = y + a \log p$

... (1)

Differentiating (1) we get,

$$\frac{dx}{dy} = 1 + \frac{a}{p} \frac{dp}{dy}$$

$$\frac{1}{p} = 1 + \frac{a}{p} \frac{dp}{dy}$$

$$\frac{1}{p} - 1 = \frac{a}{p} \frac{dp}{dy}$$

$$1 - p = a \frac{dp}{dy}$$

$$dy = a \frac{dp}{-p+1}$$

Integrating on both sides, we get

$$\int dy = -a \int \frac{dp}{p-1}$$

$$y = -a \log(p-1) + c$$

... (2)

Substituting (2) in (1) we get,

$$x = -a \log(p-1) + a \log p + c$$

$$x = a [\log p - \log(p-1)] + c$$

$$x = a \log \left(\frac{p}{(p-1)} \right) + c$$

Hence the solution is given by $x = a \log \left(\frac{p}{(p-1)} \right) + c$, $y = -a \log(p-1) + c$

20. Solve $\frac{dy}{dx} + y \cot x = \cos x$

(May -2011)

Solution: Given, $\frac{dy}{dx} + y \cot x = \cos x$

This is linear equation in y where $P = \cot x$ and $Q = \cos x$

$$I.F = e^{\int p dx} = e^{\int \cot x dx} = e^{\log \sin x}$$

$$I.F. = \sin x$$

$$ye^{\int p dx} = \int Qe^{\int p dx} dx + c$$

$$y \sin x = \int \cos x \sin x dx + c$$

$$y \sin x = \int \frac{\sin 2x}{2} dx + c$$

$$y \sin x = -\frac{\cos 2x}{4} + c$$

21. Solve $\frac{dy}{dx} + \frac{y}{x} = e^x$

Solution: Given that $\frac{dy}{dx} + \frac{y}{x} = e^x$... (1)

Equation (1) is of the form $\frac{dy}{dx} + Py = Q$ where $P = \frac{1}{x}$ & $Q = e^x$

$$\int P dx = \int \frac{dx}{x} = \log x$$

$$e^{\int P dx} = e^{\log x} = x$$

The General solution is given by

$$ye^{\int P dx} = \int Qe^{\int P dx} dx + c$$

$$yx = \int e^x dx + c$$

$$xy = (xe^x - e^x) + c$$

22. What is equation of orthogonal trajectory in Cartesian and polar co-ordinates?

Solution: The equation of orthogonal trajectory in Cartesian co-ordinates is

$$\frac{dy}{dx} = -\frac{dx}{dy}$$

The equation of orthogonal trajectory in polar co-ordinates

$$\frac{dr}{d\theta} = -r^2 \frac{d\theta}{dr}$$

23. Find the orthogonal trajectory of $xy = c^2$

Solution: Given that $xy = c^2$

Differentiating with respect to 'x', we get

$$x \frac{dy}{dx} + y = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

The equation of orthogonal trajectory is $\frac{dy}{dx} = -\frac{dx}{dy}$

$$-\frac{y}{x} = -\frac{dx}{dy}$$

$$x \, dx - y \, dy = 0$$

$$\int x \, dx - \int y \, dy = 0$$

$$\frac{x^2}{2} - \frac{y^2}{2} = c_1$$

$x^2 - y^2 = c_2$, which is the required Orthogonal trajectory.

24. What is the General solution of $y = px + ap$

Solution: The given equation is $y = px + ap$

This is clairaut's equation, put $P = c$

Then the general solution is $y = cx + ac$

25. Solve $y = (x-a) p - p^2$

Solution: Given equation is $y = (x-a) p - p^2$

This is clairaut's equation

Put $P = c$

$$y = (x-a) c - c^2 \quad \dots(1)$$

Differentiating (1) with respect to 'c'

$$0 = (x-a) - 2c$$

$$2c = (x-a)$$

$$c = \frac{x-a}{2}$$

Substituting 'c' in (1)

$$\begin{aligned} y &= (x-a) \frac{(x-a)}{2} - \frac{(x-a)^2}{4} \\ &= \frac{(x-a)^2}{2} - \frac{(x-a)^2}{4} \\ &= \frac{2(x-a)^2 - (x-a)^2}{4} = \frac{(x-a)^2}{4} \end{aligned}$$

26. Solve $p = \sin(px-y)$ where $p = \frac{dy}{dx}$

Solution: Given that $p = \sin(px-y)$

$$\sin^{-1} p = (px-y)$$

$$y = px - \sin^{-1} p \quad \dots(1)$$

Equation (1) is a Clairaut's equation

Substituting $p = c$ in equation (1), we get

$y = cx - \sin^{-1} c$ is the general solution of equation (1).

27. Solve $p = \log (px-y)$, where $p = \frac{dy}{dx}$

Solution: Given that $p = \log (px-y)$... (1)

Taking exponential on both sides.

$$e^p = e^{\log (px-y)}$$

$$e^p = px - y$$

$$y = px - e^p \quad \dots (2)$$

Equation (2) is in clairaut's form.

Sustituting $p = c$ in equation (2)

$y = cx - e^c$ is the general solution of equation (1).

28. Solve $p^2 - 9p + 18 = 0$, where $p = \frac{dy}{dx}$

Solution: Given that $p^2 - 9p + 18 = 0$
 $(p-3)(p-6) = 0$
 $p = 3$ and $p = 6$

$$\frac{dy}{dx} = 3$$

$$dy = 3dx$$

$$3dx - dy = 0$$

$$\int 3dx - \int dy = c_1$$

$$3x - y = c_1$$

$$3x - y - c_1 = 0$$

$$\begin{aligned}\frac{dy}{dx} &= 6 \\ \int 6dx - \int dy &= c_2 \\ 6x - y &= c_2 \\ 6x - y - c_2 &= 0\end{aligned}$$

The general solution is given by $(3x - y - c)(6x - y - c) = 0$

29. Define the Law of Natural Growth.

Solution: When a natural substance increasing in magnitude as a result of some action which effects all parts equally, the rate of increase depends on the amount of the substance present. For instance, if N represents the total population of a country, its rate of increase with time is $\frac{dN}{dt}$. Hence the growth of population will obey a differential equation of the

$$\text{form } \frac{dN}{dt} = kN \quad \dots(1)$$

Where k is a positive constant. More generally, if N represents any self-generating population, the rate of growth of N will be represented by the same differential equation. The solution of (1) is

$$N = N_0 e^{kt}. \text{ As } k \text{ is positive, } N \text{ is steadily increasing.}$$

30. Define the Law of Decomposition.

Solution: In a radioactive substance, atoms disintegrate spontaneously. If at any time t , the number of radioactive atoms present is N , the rate at which N is decreasing is $\frac{dN}{dt}$ and this is negative. This rate of decrease or decay is found to be proportional to itself. Hence

we have the law $\frac{dN}{dt} = -kN$ where k is a positive constant. As before, the solution of this equation is $N = N_0 e^{-kt}$ Where N_0 is the number of radioactive atoms.